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- Tiểu luận này được nhóm 1 biên soạn lại bằng phần mềm LaTeX với mục đích hoàn thành môn học thực hành Soạn thảo tài liệu khoa học bằng LaTeX, ngoài ra không còn mục đích nào khác.
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Error estimates for the finite element method of the Navier-Stokes-Poisson-Nernst-Planck equations ⋆

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ABSTRACT

In this paper, we consider the linearized backward Euler finite element scheme for the Navier-Stokes-Poisson-Nernst-Planck (NSPNP) equations. We aim to derive the unconditional error estimates of all variables in $l^\infty(L^2)$ and $l^\infty(H^1)$ norms. Compared with some previous results, we remove the mesh restriction. In addition, we derive two important physical properties for the proposed numerical scheme. Finally, three numerical examples are implemented to verify the theoretical analysis.

1. Introduction

The NSPNP equations describe the dynamics of electrically charged fluids, the motions of ionized particles or molecules and their interactions with electric fields and the surrounding fluid, which are widely used in colloidal science, biophysics, semiconductors, micro- and nanofluidics, and electrochemistry [42, 31, 46, 20]. The NSPNP equations are given as follows

$$\begin{cases} u_t - \Delta u + (u\cdot\nabla)u + \nabla p = -(c_1 - c_2)\nabla\psi, \ \text{in } \Omega\times(0,T], \\ \operatorname{div} u = 0, \ \text{in } \Omega\times(0,T], \\ c_{1t} - \Delta c_1 - \operatorname{div}(c_1\nabla\psi) + (u\cdot\nabla)c_1 = 0, \ \text{in } \Omega\times(0,T], \\ c_{2t} - \Delta c_2 + \operatorname{div}(c_2\nabla\psi) + (u\cdot\nabla)c_2 = 0, \ \text{in } \Omega\times(0,T], \\ -\Delta\psi = c_1 - c_2, \ \text{in } \Omega\times(0,T], \\ u = 0, \ \dfrac{\partial c_1}{\partial n} = \dfrac{\partial c_2}{\partial n} = \dfrac{\partial \psi}{\partial n} = 0, \ \text{on } \partial\Omega\times(0,T], \\ c_1(x,0) = c_{10}(x), \ c_2(x,0) = c_{20}(x), \ u(x,0) = u_0(x), \end{cases} \tag{1.1}$$

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where $\Omega \in \mathcal{R}^d$, $d = 2, 3$ is a bounded domain with a Lipschitz continuous boundary $\partial\Omega$, n is the outward unit-normal vector to $\partial\Omega$, c_1 and c_2 are the concentrations of positive and negative charges, u and p are the velocity and pressure of the fluid, and ψ is the quasi-electrostatic potential. For the qualitative analysis of the solutions, such as the existence, uniqueness, stability and asymptotic behavior, we refer to [47, 29, 12, 1, 6, 5, 19].

Many numerical methods have been developed for the PNP and NSPNP equations. It is crucial to preserve some important physical properties at the discrete level, such as the mass conservation, concentration positivity and energy stability, and the related work can be referred to [13, 22, 39, 37, 11, 43, 14, 44, 45, 52, 40, 48, 28, 18, 7, 17]. In addition, some numerical techniques and implementation strategies were proposed to improve the performance efficiency, such as the adaptive finite element methods [21, 58, 60], the two grid methods [49], the stabilized techniques [55, 32], the proper orthogonal decomposition method [9], and some improved iteration techniques [59, 56, 38, 2].

There are some studies about the error estimates of the finite element method for the PNP and NSPNP equations. Sun and He analyzed an implicit Crank-Nicolson finite element scheme for the PNP equations in [53], and obtained the optimal error estimates in $l^\infty(H^1)$ norm for all variables, however, the convergence order of $l^\infty(L^2)$ norm was suboptimal. They also proposed the mixed finite element method for the PNP equations [23], and obtained optimal error estimates for $l^\infty(H^1)$ and $l^\infty(L^2)$ norms, and later they discussed the PNP-Stokes equations and modified NSPNP equations [24, 25]. Shi and Yang et al. investigated the conforming and nonconforming finite element methods on rectangular meshes [50, 51], and obtained the superconvergence results by integral identities techniques. Yang et al. researched the virtual element method on general polygonal meshes [57], and derived the error estimates in $l^\infty(H^1)$ and $l^\infty(L^2)$ norms. However, due to the use of inverse inequality and mathematical inductive argument, a mesh restriction is needed in the above references [53, 23, 24, 25, 50, 51, 57]. Gao and He [15, 16] presented linearized conservative finite element and mixed finite element methods for the PNP equations, and established unconditionally optimal error estimates for all variables in $l^2(H^1)$ and $l^\infty(L^2)$ norms, but they do not derive the error estimates in $l^\infty(H^1)$ norm. For the convergence analysis and error estimates of some other gradient flow model coupled with fluid motion, such as the Cahn-Hilliard-Hele-Shaw equations and the Cahn-Hilliard-Navier-Stokes system, one can refer to [4, 3, 41, 10].

In this paper, we focus on the error estimates of the linearized backward Euler finite element scheme for the NSPNP equations, and use low order conforming finite element pairs. Firstly, under an inductive hypothesis, we derive the existence and uniqueness of solutions, and the unconditional error estimates of all variables in $l^\infty(L^2)$ norms. Then, we obtain the $l^\infty(H^1)$ and $l^2(H^2)$ norm stability of the finite element solutions. Next, based on the stability, we deduce the unconditional error estimates in $l^\infty(H^1)$ norm. Moreover, the global mass conservation and the energy stability are also derived for the proposed numerical scheme. Finally, several numerical examples are implemented to verify the theoretical analysis. Compared with the results in [53, 23, 24, 25, 50, 51, 57], we remove the mesh restriction.

Throughout the article, the notations of the standard Sobolev spaces are used. The generic positive constant C with or without subscript is independent of the mesh size h and the time step τ.

2. Preliminary

We introduce the following function spaces

$$X = H^1(\Omega), \quad X^0 = \Big\{ \eta \in H^1(\Omega) : \int_\Omega \eta \, \mathrm{d}x = 0 \Big\}, \tag{2.1}$$

$$V = \big(H_0^1(\Omega)\big)^d, \quad Q = L_0^2(\Omega) = \Big\{ q \in L^2(\Omega) : \int_\Omega q \, \mathrm{d}x = 0 \Big\}. \tag{2.2}$$

The trilinear forms $b_1(\cdot\,;\cdot,\cdot)$ on $V \times V \times V$ and $b_2(\cdot\,;\cdot,\cdot)$ on $V \times X \times X$ are defined by

$$b_1(u;\, v,\, w) = \tfrac{1}{2}\big((u \cdot \nabla)v,\, w\big) - \tfrac{1}{2}\big((u \cdot \nabla)w,\, v\big), \quad \forall\, u, v, w \in V, \tag{2.3}$$

$$b_2(u;\, c,\, \eta) = \tfrac{1}{2}(u \cdot \nabla c,\, \eta) - \tfrac{1}{2}(u \cdot \nabla \eta,\, c), \quad \forall\, u \in V,\ c, \eta \in X. \tag{2.4}$$

We recall the following Poincaré–Friedrichs inequalities

$$\|v\|_1 \le C\, \|\nabla v\|_0, \quad \forall\, v \in V, \tag{2.5}$$

$$\|\varphi\|_1 \le C\Big(\|\nabla \varphi\|_0 + \Big| \int_\Omega \varphi \, \mathrm{d}x \Big| \Big), \quad \forall\, \varphi \in X, \tag{2.6}$$

and the embedding inequalities

$$\|\varphi\|_{0,6} \le C\,\|\varphi\|_1, \quad \forall\,\varphi \in X. \tag{2.7}$$

Next, integrating the third equation of (**??**) and using the Green formula, we can derive

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\int_\Omega c_1\,\mathrm{d}x\right) - \int_{\partial\Omega} \frac{\partial c_1}{\partial n}\,\mathrm{d}s - \int_{\partial\Omega} c_1 \frac{\partial \psi}{\partial n}\,\mathrm{d}s + \int_{\partial\Omega} u\cdot n\, c_1\,\mathrm{d}s = 0. \tag{2.8}$$

Then in view of the boundary conditions, we get

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\int_\Omega c_1\,\mathrm{d}x\right) = 0, \tag{2.9}$$

which implies

$$\int_\Omega c_1\,\mathrm{d}x = \int_\Omega c_{10}\,\mathrm{d}x. \tag{2.10}$$

Manipulating the fourth and fifth equations of (**??**) in the same way, we obtain

$$\int_\Omega c_2\,\mathrm{d}x = \int_\Omega c_{20}\,\mathrm{d}x, \qquad \int_\Omega c_1\,\mathrm{d}x = \int_\Omega c_2\,\mathrm{d}x. \tag{2.11}$$

Set

$$m_0 = \frac{1}{|\Omega|}\int_\Omega c_{10}\,\mathrm{d}x = \frac{1}{|\Omega|}\int_\Omega c_{20}\,\mathrm{d}x. \tag{2.12}$$

Take $\tilde{c}_1 = c_1 - m_0$, $\tilde{c}_2 = c_2 - m_0$; then we obtain the variational form of equations (**??**): find $(u, p, \tilde{c}_1, \tilde{c}_2, \psi) \in V \times Q \times X^0 \times X^0 \times X^0$ such that

$$\begin{cases} (u_t, v) + (\nabla u, \nabla v) + b_1(u; u, v) - (\operatorname{div} v,\, p) = -\big((\tilde{c}_1 - \tilde{c}_2)\nabla\psi,\, v\big), & \forall\, v \in V, \\ (\operatorname{div} u,\, q) = 0, & \forall\, q \in Q, \\ (\tilde{c}_{1t},\, \eta_1) + (\nabla\tilde{c}_1, \nabla\eta_1) + \big((\tilde{c}_1 + m_0)\nabla\psi,\, \nabla\eta_1\big) + b_2(u; \tilde{c}_1, \eta_1) = 0, & \forall\, \eta_1 \in X, \\ (\tilde{c}_{2t},\, \eta_2) + (\nabla\tilde{c}_2, \nabla\eta_2) - \big((\tilde{c}_2 + m_0)\nabla\psi,\, \nabla\eta_2\big) + b_2(u; \tilde{c}_2, \eta_2) = 0, & \forall\, \eta_2 \in X, \\ (\nabla\psi, \nabla\varphi) = (\tilde{c}_1 - \tilde{c}_2,\, \varphi), & \forall\, \varphi \in X. \end{cases} \tag{2.13}$$

Let $\mathcal{T}_h$ be a regular and quasi-uniform triangular or tetrahedral partition of Ω. We consider low-order conforming finite element spaces $V_h \times Q_h \times X_h^0 \times X_h^0 \times X_h^0 \subset V \times Q \times X^0 \times X^0 \times X^0$, which satisfy the following approximation properties: for $u, \tilde{c}_1, \tilde{c}_2, \psi \in H^2(\Omega)$, $p \in H^1(\Omega)$, there exist $v_h, q_h, \chi_{1h}, \chi_{2h}, \varphi_h \in V_h \times Q_h \times X_h^0 \times X_h^0 \times X_h^0$ such that

$$\|\tilde{c}_i - \chi_{ih}\|_0 + h\,\|\nabla(\tilde{c}_i - \chi_{ih})\|_0 \le Ch^2\|\tilde{c}_i\|_2, \quad i = 1, 2, \tag{2.14}$$

$$\|\psi - \varphi_h\|_0 + h\,\|\nabla(\psi - \varphi_h)\|_0 \le Ch^2\|\psi\|_2, \tag{2.15}$$

$$\|u - v_h\|_0 + h\big(\|\nabla(u - v_h)\|_0 + \|p - q_h\|_0\big) \le Ch^2\big(\|u\|_2 + \|p\|_1\big), \tag{2.16}$$

and the inf-sup condition [30]

$$\sup_{v_h \in V_h} \frac{(\operatorname{div} v_h,\, q_h)}{\|\nabla v_h\|_0} \ge C\,\|q_h\|_0, \quad \forall\, q_h \in Q_h. \tag{2.17}$$

We define the Ritz projection $R_h : X^0 \to X_h^0$ and the Stokes projection $S_h : V \times Q \to V_h \times Q_h$ by

$$\big(\nabla(R_h\eta - \eta),\, \nabla\chi_h\big) = 0, \quad \forall\, \chi_h \in X_h^0, \tag{2.18}$$

and

$$\begin{cases} (\nabla S_h u, \nabla v_h) - (\operatorname{div} v_h, S_h p) = (\nabla u, \nabla v_h) - (\operatorname{div} v_h, p), & \forall\, v_h \in V_h, \\ (\operatorname{div} S_h u,\ q_h) = 0, & \forall\, q_h \in Q_h. \end{cases} \tag{2.19}$$

Then, from the standard finite element theory [30, 36], we have

$$\|\eta - R_h\eta\|_0 + h\,\|\nabla(\eta - R_h\eta)\|_0 \le Ch^2\|\eta\|_2, \tag{2.20}$$

$$\|u - S_h u\|_0 + h\big(\|\nabla(u - S_h u)\|_0 + \|p - S_h p\|_0\big) \le Ch^2\big(\|u\|_2 + \|p\|_1\big), \tag{2.21}$$

$$\|u_t - S_h u_t\|_0 \le Ch^2\big(\|u_t\|_2 + \|p_t\|_1\big). \tag{2.22}$$

The discrete operator $\Delta_h : X_h^0 \to X_h^0$ is given as follows

$$(\Delta_h \eta_h, \chi_h) = (\nabla\eta_h, \nabla\chi_h), \quad \forall\, \chi_h \in X_h^0. \tag{2.23}$$

The subspace V_h^{div} of V_h is introduced as

$$V_h^{\mathrm{div}} = \big\{v_h \in V_h : (\operatorname{div} v_h, q_h) = 0,\ \forall\, q_h \in Q_h\big\}. \tag{2.24}$$

And the discrete Stokes operator $A_h : V_h^{\mathrm{div}} \to V_h^{\mathrm{div}}$ is defined by

$$(A_h u_h, v_h) = (\nabla u_h, \nabla v_h), \quad \forall\, v_h \in V_h^{\mathrm{div}}. \tag{2.25}$$

Then, we have the following Lemmas 2.1–2.4, which can be found in [41, 10, 26, 27], or proven similarly.

Lemma 2.1. *For $\chi_h \in X_h^0$, there holds*

$$\|\chi_h\|_{0,\infty} + \|\nabla\chi_h\|_{0,6} \le C\,\|\Delta_h\chi_h\|_0. \tag{2.26}$$

Lemma 2.2. *For $u_h, v_h, w_h \in V_h^{\mathrm{div}}$, there hold*

$$b_1(u_h; v_h, w_h) \le C\,\|\nabla u_h\|_0\|\nabla v_h\|_0\|\nabla w_h\|_0, \tag{2.27}$$
$$b_1(u_h; v_h, w_h) \le C\,\|u_h\|_0\|\nabla v_h\|_0^{1/2}\|A_h v_h\|_0^{1/2}\|\nabla w_h\|_0, \tag{2.28}$$
$$b_1(u_h; v_h, w_h) \le C\,\|\nabla u_h\|_0\|\nabla v_h\|_0^{1/2}\|A_h v_h\|_0^{1/2}\|w_h\|_0, \tag{2.29}$$
$$b_1(u_h; v_h, w_h) \le C\,\|\nabla u_h\|_0^{1/2}\|A_h u_h\|_0^{1/2}\|\nabla v_h\|_0\|w_h\|_0, \tag{2.30}$$
$$b_1(u_h; v_h, w_h) \le C\,\|u_h\|_0\|\nabla u_h\|_0^{1/2}\|A_h v_h\|_0\|w_h\|_0. \tag{2.31}$$

Lemma 2.3. *For $u_h \in V_h^{\mathrm{div}}$, $\varphi_h, \chi_h \in X_h^0$, there hold*

$$b_2(u_h; \varphi_h, \chi_h) \le C\,\|\nabla u_h\|_0\|\nabla\chi_h\|_0\|\nabla\varphi_h\|_0, \tag{2.32}$$
$$b_2(u_h; \varphi_h, \chi_h) \le C\,\|u_h\|_0\|\nabla\varphi_h\|_0^{1/2}\|\Delta_h\varphi_h\|_0^{1/2}\|\nabla\chi_h\|_0, \tag{2.33}$$
$$b_2(u_h; \varphi_h, \chi_h) \le C\,\|\nabla u_h\|_0^{1/2}\|A_h u_h\|_0^{1/2}\|\nabla\varphi_h\|_0\|\chi_h\|_0, \tag{2.34}$$
$$b_2(u_h; \varphi_h, \chi_h) \le C\,\|\nabla u_h\|_0\|\nabla\varphi_h\|_0^{1/2}\|\Delta_h\varphi_h\|_0^{1/2}\|\chi_h\|_0. \tag{2.35}$$

Lemma 2.4. *For $\eta_h, \varphi_h, \chi_h \in X_h^0$, there holds*

$$(\eta_h\nabla\varphi_h, \nabla\chi_h) \le C\,\|\nabla\eta_h\|_0^{1/2}\|\Delta_h\eta_h\|_0^{1/2}\|\Delta_h\varphi_h\|_0\|\chi_h\|_0. \tag{2.36}$$

Finally, assume that the solutions and initial values of equations (**??**) satisfy the following regularity condition:

$$\begin{aligned} &\|u_0\|_2^2 + \|c_{10}\|_2^2 + \|c_{20}\|_2^2 + \|c_1\|_{L^\infty(H^2)}^2 + \|c_2\|_{L^\infty(H^2)}^2 + \|\psi\|_{L^\infty(H^3)}^2 \\ &\quad + \|u\|_{L^\infty(H^2)}^2 + \|p\|_{L^\infty(H^1)}^2 + \|c_{1t}\|_{L^2(H^2)}^2 + \|c_{1tt}\|_{L^2(L^2)}^2 + \|c_{2t}\|_{L^2(H^2)}^2 \\ &\quad + \|c_{2tt}\|_{L^2(L^2)}^2 + \|u_t\|_{L^2(H^2)}^2 + \|u_{tt}\|_{L^2(L^2)}^2 + \|p_t\|_{L^2(H^1)}^2 \le C_0. \end{aligned} \tag{2.37}$$

3. The linearized backward Euler scheme

Let $\{t_j : t_j = j\tau,\ 0 \le j \le J\}$ be a uniform partition of the time interval $[0,T]$ with time step $\tau = T/J$. Denote $f^j = f(\cdot, t_j)$, and for a sequence of functions $\{f^j\}_{j=0}^J$ we define

$$D_\tau f^{j+1} = \frac{f^{j+1} - f^j}{\tau}. \tag{3.1}$$

With the above notation, the linearized backward Euler scheme of (2.13) is given as follows.

Step 1. Set $(\tilde{c}_{1h}^0, \tilde{c}_{2h}^0, u_h^0) = (R_h\tilde{c}_{10}, R_h\tilde{c}_{20}, S_h u_0)$. For $j = 0, 1, \dots, J-1$, find $(u_h^{j+1}, p_h^{j+1}, \tilde{c}_{1h}^{j+1}, \tilde{c}_{2h}^{j+1}, \psi_h^{j+1}) \in V_h \times Q_h \times X_h^0 \times X_h^0 \times X_h^0$ such that

$$\begin{cases} \big(D_\tau u_h^{j+1}, v_h\big) + \big(\nabla u_h^{j+1}, \nabla v_h\big) + b_1\big(u_h^j;\, u_h^{j+1},\, v_h\big) - \big(\operatorname{div} v_h,\, p_h^{j+1}\big) \\ \quad = -\big((\tilde{c}_{1h}^j - \tilde{c}_{2h}^j)\nabla\psi_h^{j+1},\, v_h\big), \quad \forall\, v_h \in V_h, \\ \big(\operatorname{div} u_h^{j+1},\, q_h\big) = 0, \quad \forall\, q_h \in Q_h, \\ \big(D_\tau \tilde{c}_{1h}^{j+1},\, \eta_{1h}\big) + \big(\nabla \tilde{c}_{1h}^{j+1}, \nabla\eta_{1h}\big) + \big((\tilde{c}_{1h}^j + m_0)\nabla\psi_h^{j+1}, \nabla\eta_{1h}\big) \\ \quad + b_2\big(u_h^j;\, \tilde{c}_{1h}^{j+1},\, \eta_{1h}\big) = 0, \quad \forall\, \eta_{1h} \in X_h, \\ \big(D_\tau \tilde{c}_{2h}^{j+1},\, \eta_{2h}\big) + \big(\nabla \tilde{c}_{2h}^{j+1}, \nabla\eta_{2h}\big) - \big((\tilde{c}_{2h}^j + m_0)\nabla\psi_h^{j+1}, \nabla\eta_{2h}\big) \\ \quad + b_2\big(u_h^j;\, \tilde{c}_{2h}^{j+1},\, \eta_{2h}\big) = 0, \quad \forall\, \eta_{2h} \in X_h, \\ \big(\nabla\psi_h^{j+1}, \nabla\varphi_h\big) = \big(\tilde{c}_{1h}^{j+1} - \tilde{c}_{2h}^{j+1},\, \varphi_h\big), \quad \forall\, \varphi_h \in X_h. \end{cases} \tag{3.2}$$

Step 2. Set

$$c_{1h}^{j+1} = \tilde{c}_{1h}^{j+1} + m_0, \qquad c_{2h}^{j+1} = \tilde{c}_{2h}^{j+1} + m_0. \tag{3.3}$$

To derive the well-posedness and unconditional error estimates of scheme (3.2), we need the following inductive hypothesis:

$$\left\|\tilde{c}_{1h}^j\right\|_{0,3} + \left\|\tilde{c}_{2h}^j\right\|_{0,3} \le C_1. \tag{3.4}$$

In the remainder of this section we prove the existence and uniqueness of scheme (3.2).

Theorem 3.1. *If the time step τ is small enough, scheme* (3.2) *has a unique solution.*

Proof. Since scheme (3.2) is an algebraic linear system, we only need to prove uniqueness. Suppose there exist $(\tilde{c}_{1h,i}^{j+1}, \tilde{c}_{2h,i}^{j+1}, \psi_{h,i}^{j+1}) \in X_h^0 \times X_h^0 \times X_h^0$, $i = 1, 2$, satisfying the last three equations of (3.2). Then we have

$$\begin{cases} \big(D_\tau(\tilde{c}_{1h,1}^{j+1} - \tilde{c}_{1h,2}^{j+1}),\, \eta_{1h}\big) + \big(\nabla(\tilde{c}_{1h,1}^{j+1} - \tilde{c}_{1h,2}^{j+1}), \nabla\eta_{1h}\big) \\ \quad + \big((\tilde{c}_{1h}^j + m_0)\nabla(\psi_{h,1}^{j+1} - \psi_{h,2}^{j+1}), \nabla\eta_{1h}\big) + b_2\big(u_h^j;\, \tilde{c}_{1h,1}^{j+1} - \tilde{c}_{1h,2}^{j+1},\, \eta_{1h}\big) = 0, \\ \big(D_\tau(\tilde{c}_{2h,1}^{j+1} - \tilde{c}_{2h,2}^{j+1}),\, \eta_{2h}\big) + \big(\nabla(\tilde{c}_{2h,1}^{j+1} - \tilde{c}_{2h,2}^{j+1}), \nabla\eta_{2h}\big) \\ \quad - \big((\tilde{c}_{2h}^j + m_0)\nabla(\psi_{h,1}^{j+1} - \psi_{h,2}^{j+1}), \nabla\eta_{2h}\big) + b_2\big(u_h^j;\, \tilde{c}_{2h,1}^{j+1} - \tilde{c}_{2h,2}^{j+1},\, \eta_{2h}\big) = 0, \\ \big(\nabla(\psi_{h,1}^{j+1} - \psi_{h,2}^{j+1}), \nabla\varphi_h\big) = \big((\tilde{c}_{1h,1}^{j+1} - \tilde{c}_{1h,2}^{j+1}) - (\tilde{c}_{2h,1}^{j+1} - \tilde{c}_{2h,2}^{j+1}),\, \varphi_h\big). \end{cases} \tag{3.5}$$

Taking $\varphi_h = \psi_{h,1}^{j+1} - \psi_{h,2}^{j+1}$ and $\varphi_h = \Delta_h(\psi_{h,1}^{j+1} - \psi_{h,2}^{j+1})$ in the third equation of (3.5), respectively, we obtain

$$\left\|\nabla(\psi_{h,1}^{j+1} - \psi_{h,2}^{j+1})\right\|_0 \le C\Big(\left\|\tilde{c}_{1h,1}^{j+1} - \tilde{c}_{1h,2}^{j+1}\right\|_0 + \left\|\tilde{c}_{2h,1}^{j+1} - \tilde{c}_{2h,2}^{j+1}\right\|_0\Big), \tag{3.6}$$

$$\left\|\Delta_h(\psi_{h,1}^{j+1} - \psi_{h,2}^{j+1})\right\|_0 \le \left\|\tilde{c}_{1h,1}^{j+1} - \tilde{c}_{1h,2}^{j+1}\right\|_0 + \left\|\tilde{c}_{2h,1}^{j+1} - \tilde{c}_{2h,2}^{j+1}\right\|_0. \tag{3.7}$$

Setting $\eta_{1h} = \tilde{c}_{1h,1}^{j+1} - \tilde{c}_{1h,2}^{j+1}$ in the first equation of (3.5) and using (2.26), (3.4), (3.7), we arrive at

$$\left\|\tilde{c}_{1h,1}^{j+1} - \tilde{c}_{1h,2}^{j+1}\right\|_0^2 + \tfrac{\tau}{2}\left\|\nabla(\tilde{c}_{1h,1}^{j+1} - \tilde{c}_{1h,2}^{j+1})\right\|_0^2 \le CC_1^2\tau\Big(\left\|\tilde{c}_{1h,1}^{j+1} - \tilde{c}_{1h,2}^{j+1}\right\|_0^2 + \left\|\tilde{c}_{2h,1}^{j+1} - \tilde{c}_{2h,2}^{j+1}\right\|_0^2\Big). \tag{3.8}$$

Similarly, we get

$$\left\|\tilde{c}_{2h,1}^{j+1}-\tilde{c}_{2h,2}^{j+1}\right\|_0^2+\tfrac{\tau}{2}\left\|\nabla(\tilde{c}_{2h,1}^{j+1}-\tilde{c}_{2h,2}^{j+1})\right\|_0^2\le CC_1^2\tau\big(\left\|\tilde{c}_{1h,1}^{j+1}-\tilde{c}_{1h,2}^{j+1}\right\|_0^2+\left\|\tilde{c}_{2h,1}^{j+1}-\tilde{c}_{2h,2}^{j+1}\right\|_0^2\big). \tag{3.9}$$

Summing (3.8) and (3.9) gives

$$\begin{aligned}&(1-CC_1^2\tau)\big(\left\|\tilde{c}_{1h,1}^{j+1}-\tilde{c}_{1h,2}^{j+1}\right\|_0^2+\left\|\tilde{c}_{2h,1}^{j+1}-\tilde{c}_{2h,2}^{j+1}\right\|_0^2\big)\\&\quad+\tfrac{\tau}{2}\big(\left\|\nabla(\tilde{c}_{1h,1}^{j+1}-\tilde{c}_{1h,2}^{j+1})\right\|_0^2+\left\|\nabla(\tilde{c}_{2h,1}^{j+1}-\tilde{c}_{2h,2}^{j+1})\right\|_0^2\big)\le 0.\end{aligned} \tag{3.10}$$

Setting $\tau<\tau_1:=\dfrac{1}{2CC_1^2}$ in (3.10) yields

$$\tilde{c}_{1h,1}^{j+1}=\tilde{c}_{1h,2}^{j+1},\qquad \tilde{c}_{2h,1}^{j+1}=\tilde{c}_{2h,2}^{j+1}, \tag{3.11}$$

which together with (3.6) leads to

$$\psi_{h,1}^{j+1}=\psi_{h,2}^{j+1}. \tag{3.12}$$

Finally, the first two equations of (3.2) represent the linearized backward Euler scheme of the Navier–Stokes equations, which has the unique solution (u_h^{j+1},p_h^{j+1}). The proof is completed. □

4. The error estimates

In this section, we will derive the unconditional error estimates of all the variables. Let

$$\begin{aligned}&e_u^{j+1}=u_h^{j+1}-S_hu^{j+1},\quad e_p^{j+1}=p_h^{j+1}-S_hp^{j+1},\\&e_{\tilde{c}_1}^{j+1}=\tilde{c}_{1h}^{j+1}-R_h\tilde{c}_1^{j+1},\quad e_{\tilde{c}_2}^{j+1}=\tilde{c}_{2h}^{j+1}-R_h\tilde{c}_2^{j+1},\quad e_\psi^{j+1}=\psi_h^{j+1}-R_h\psi^{j+1},\end{aligned} \tag{4.1}$$

and subtract (3.2) from (2.13) with $t=t_{j+1}$, then we obtain the error equations

$$\begin{aligned}&\big(u_t^{j+1}-D_\tau u^{j+1},v_h\big)+\big(D_\tau(u^{j+1}-S_hu^{j+1}),v_h\big)-\big(D_\tau e_u^{j+1},v_h\big)-\big(\nabla e_u^{j+1},\nabla v_h\big)\\&\quad+b_1(u^{j+1}-u^j;u^{j+1},v_h)+b_1(u^j-S_hu^j;u^{j+1},v_h)+b_1(S_hu^j;u^{j+1}-S_hu^{j+1},v_h)\\&\quad-b_1(e_u^j;S_hu^{j+1},v_h)-b_1(u_h^j;e_u^{j+1},v_h)+\big(e_p^{j+1},\operatorname{div}v_h\big)\\&\quad+\big(\tilde{c}_1^{j+1}\nabla\psi^{j+1}-\tilde{c}_{1h}^j\nabla\psi_h^{j+1},v_h\big)-\big(\tilde{c}_2^{j+1}\nabla\psi^{j+1}-\tilde{c}_{2h}^j\nabla\psi_h^{j+1},v_h\big)=0,\quad\forall v_h\in V_h,\end{aligned} \tag{4.2}$$

$$\big(\operatorname{div}e_u^{j+1},q_h\big)=0,\quad\forall q_h\in Q_h, \tag{4.3}$$

$$\begin{aligned}&\big(\tilde{c}_{1t}^{j+1}-D_\tau\tilde{c}_1^{j+1},\eta_{1h}\big)+\big(D_\tau\tilde{c}_1^{j+1}-D_\tau R_h\tilde{c}_1^{j+1},\eta_{1h}\big)-\big(D_\tau e_{\tilde{c}_1}^{j+1},\eta_{1h}\big)-\big(\nabla e_{\tilde{c}_1}^{j+1},\nabla\eta_{1h}\big)\\&\quad-m_0\big(\nabla e_\psi^{j+1},\nabla\eta_{1h}\big)+\big((\tilde{c}_1^{j+1}-\tilde{c}_1^j)\nabla\psi^{j+1},\nabla\eta_{1h}\big)+\big((\tilde{c}_1^j-R_h\tilde{c}_1^j)\nabla\psi^{j+1},\nabla\eta_{1h}\big)\\&\quad-\big(e_{\tilde{c}_1}^j\nabla\psi^{j+1},\nabla\eta_{1h}\big)+\big(\tilde{c}_{1h}^j\nabla(\psi^{j+1}-R_h\psi^{j+1}),\nabla\eta_{1h}\big)-\big(\tilde{c}_{1h}^j\nabla e_\psi^{j+1},\nabla\eta_{1h}\big)\\&\quad+b_2(u^{j+1}-u^j;\tilde{c}_1^{j+1},\eta_{1h})+b_2(u^j-S_hu^j;\tilde{c}_1^{j+1},\eta_{1h})-b_2(e_u^j;\tilde{c}_1^{j+1},\eta_{1h})\\&\quad+b_2(u_h^j;\tilde{c}_1^{j+1}-R_h\tilde{c}_1^{j+1},\eta_{1h})-b_2(u_h^j;e_{\tilde{c}_1}^{j+1},\eta_{1h})=0,\quad\forall\eta_{1h}\in X_h,\end{aligned} \tag{4.4}$$

$$\begin{aligned}&\big(\tilde{c}_{2t}^{j+1}-D_\tau\tilde{c}_2^{j+1},\eta_{2h}\big)+\big(D_\tau\tilde{c}_2^{j+1}-D_\tau R_h\tilde{c}_2^{j+1},\eta_{2h}\big)-\big(D_\tau e_{\tilde{c}_2}^{j+1},\eta_{2h}\big)-\big(\nabla e_{\tilde{c}_2}^{j+1},\nabla\eta_{2h}\big)\\&\quad+m_0\big(\nabla e_\psi^{j+1},\nabla\eta_{2h}\big)-\big((\tilde{c}_2^{j+1}-\tilde{c}_2^j)\nabla\psi^{j+1},\nabla\eta_{2h}\big)-\big((\tilde{c}_2^j-R_h\tilde{c}_2^j)\nabla\psi^{j+1},\nabla\eta_{2h}\big)\\&\quad+\big(e_{\tilde{c}_2}^j\nabla\psi^{j+1},\nabla\eta_{2h}\big)-\big(\tilde{c}_{2h}^j\nabla(\psi^{j+1}-R_h\psi^{j+1}),\nabla\eta_{2h}\big)+\big(\tilde{c}_{2h}^j\nabla e_\psi^{j+1},\nabla\eta_{2h}\big)\\&\quad+b_2(u^{j+1}-u^j;\tilde{c}_2^{j+1},\eta_{2h})+b_2(u^j-S_hu^j;\tilde{c}_2^{j+1},\eta_{2h})-b_2(e_u^j;\tilde{c}_2^{j+1},\eta_{2h})\\&\quad+b_2(u_h^j;\tilde{c}_2^{j+1}-R_h\tilde{c}_2^{j+1},\eta_{2h})-b_2(u_h^j;e_{\tilde{c}_2}^{j+1},\eta_{2h})=0,\quad\forall\eta_{2h}\in X_h,\end{aligned} \tag{4.5}$$

$$\big(\nabla e_\psi^{j+1},\nabla\varphi_h\big)=-\big(\tilde{c}_1^{j+1}-R_h\tilde{c}_1^{j+1},\varphi_h\big)+\big(\tilde{c}_2^{j+1}-R_h\tilde{c}_2^{j+1},\varphi_h\big)+\big(e_{\tilde{c}_1}^{j+1}-e_{\tilde{c}_2}^{j+1},\varphi_h\big),\quad\forall\varphi_h\in X_h. \tag{4.6}$$

4.1. The $\ell^\infty(L^2)$-norm error estimates

Now, we derive the $\ell^\infty(L^2)$-norm error estimates of all variables, which are shown in following Lemmas 4.1–4.3 and Theorem 4.4.

Lemma 4.1. *Assume that* $(\tilde{c}_1, \tilde{c}_2, \psi, u, p)$ *and* $(\tilde{c}_{1h}^{j+1}, \tilde{c}_{2h}^{j+1}, \psi_h^{j+1}, u_h^{j+1}, p_h^{j+1})$ *are solutions of* (2.13) *and* (3.2)*, respectively, and the regularity condition* (2.37) *holds. Then we have*

$$\left\|\nabla e_\psi^{j+1}\right\|_0 + \left\|\Delta_h e_\psi^{j+1}\right\|_0 \le CC_0h^2 + C\Big(\left\|e_{\tilde{c}_1}^{j+1}\right\|_0 + \left\|e_{\tilde{c}_2}^{j+1}\right\|_0\Big). \tag{4.7}$$

Proof. Taking $\varphi_h = e_\psi^{j+1}$ and $\varphi_h = \Delta_h e_\psi^{j+1}$ in (4.6), respectively, we get

$$\begin{aligned}\left\|\nabla e_\psi^{j+1}\right\|_0^2 \le{}& C\Big(\left\|\tilde{c}_1^{j+1} - R_h\tilde{c}_1^{j+1}\right\|_0 + \left\|\tilde{c}_2^{j+1} - R_h\tilde{c}_2^{j+1}\right\|_0\Big)\left\|\nabla e_\psi^{j+1}\right\|_0\\ &+ C\Big(\left\|e_{\tilde{c}_1}^{j+1}\right\|_0 + \left\|e_{\tilde{c}_2}^{j+1}\right\|_0\Big)\left\|\nabla e_\psi^{j+1}\right\|_0,\end{aligned} \tag{4.8}$$

$$\begin{aligned}\left\|\Delta_h e_\psi^{j+1}\right\|_0^2 \le{}& \Big(\left\|\tilde{c}_1^{j+1} - R_h\tilde{c}_1^{j+1}\right\|_0 + \left\|\tilde{c}_2^{j+1} - R_h\tilde{c}_2^{j+1}\right\|_0\Big)\left\|\Delta_h e_\psi^{j+1}\right\|_0\\ &+ \Big(\left\|e_{\tilde{c}_1}^{j+1}\right\|_0 + \left\|e_{\tilde{c}_2}^{j+1}\right\|_0\Big)\left\|\Delta_h e_\psi^{j+1}\right\|_0,\end{aligned} \tag{4.9}$$

which implies the desired result (4.7). □

Lemma 4.2. *Under the conditions of Lemma 4.1, we have*

$$\begin{aligned}&\left\|e_{\tilde{c}_1}^J\right\|_0^2 + \left\|e_{\tilde{c}_2}^J\right\|_0^2 + \tau\sum_{j=0}^{J-1}\Big(\left\|\nabla e_{\tilde{c}_1}^{j+1}\right\|_0^2 + \left\|\nabla e_{\tilde{c}_2}^{j+1}\right\|_0^2\Big)\\ &\quad\le CC_0^2(C_0^2 + C_1^2)(\tau^2 + h^2) + C(C_0^2 + C_1^2)\,\tau\sum_{j=0}^{J-1}\Big(\left\|e_{\tilde{c}_1}^{j+1}\right\|_0^2 + \left\|e_{\tilde{c}_2}^{j+1}\right\|_0^2 + \left\|e_u^{j+1}\right\|_0^2\Big).\end{aligned} \tag{4.10}$$

Proof. Setting $\eta_{1h} = e_{\tilde{c}_1}^{j+1}$ in (4.4) leads to

$$\begin{aligned}&\frac{1}{2\tau}\Big(\left\|e_{\tilde{c}_1}^{j+1}\right\|_0^2 + \left\|e_{\tilde{c}_1}^{j+1} - e_{\tilde{c}_1}^j\right\|_0^2 - \left\|e_{\tilde{c}_1}^j\right\|_0^2\Big) + \left\|\nabla e_{\tilde{c}_1}^{j+1}\right\|_0^2\\ &= \big(\tilde{c}_{1t}^{j+1} - D_\tau\tilde{c}_1^{j+1},\, e_{\tilde{c}_1}^{j+1}\big) + \big(D_\tau\tilde{c}_1^{j+1} - D_\tau R_h\tilde{c}_1^{j+1},\, e_{\tilde{c}_1}^{j+1}\big) - m_0\big(\nabla e_\psi^{j+1},\, \nabla e_{\tilde{c}_1}^{j+1}\big)\\ &\quad + \big((\tilde{c}_1^{j+1} - \tilde{c}_1^j)\nabla\psi^{j+1},\, \nabla e_{\tilde{c}_1}^{j+1}\big) + \big((\tilde{c}_1^j - R_h\tilde{c}_1^j)\nabla\psi^{j+1},\, \nabla e_{\tilde{c}_1}^{j+1}\big) - \big(e_{\tilde{c}_1}^j\nabla\psi^{j+1},\, \nabla e_{\tilde{c}_1}^{j+1}\big)\\ &\quad + \big(\tilde{c}_{1h}^j\nabla(\psi^{j+1} - R_h\psi^{j+1}),\, \nabla e_{\tilde{c}_1}^{j+1}\big) - \big(\tilde{c}_{1h}^j\nabla e_\psi^{j+1},\, \nabla e_{\tilde{c}_1}^{j+1}\big)\\ &\quad + b_2(u^{j+1} - u^j;\, \tilde{c}_1^{j+1},\, e_{\tilde{c}_1}^{j+1}) + b_2(u^j - S_hu^j;\, \tilde{c}_1^{j+1},\, e_{\tilde{c}_1}^{j+1}) - b_2(e_u^j;\, \tilde{c}_1^{j+1},\, e_{\tilde{c}_1}^{j+1})\\ &\quad + b_2(u_h^j;\, \tilde{c}_1^{j+1} - R_h\tilde{c}_1^{j+1},\, e_{\tilde{c}_1}^{j+1}) \;:=\; \sum_{i=1}^{12} E_i.\end{aligned} \tag{4.11}$$

Now, we bound the above terms. From the Taylor formula, Lemma 4.1 and (**??**), it is not hard to check that

$$\begin{aligned}\sum_{i=1}^{3}|E_i| \le{}& C\tau\int_{t_j}^{t_{j+1}}\|\tilde{c}_{1tt}\|_0^2\,\mathrm{d}t + \frac{Ch^4}{\tau}\int_{t_j}^{t_{j+1}}\|\tilde{c}_{1t}\|_2^2\,\mathrm{d}t + CC_0^2h^4\\ &+ C\Big(\left\|e_{\tilde{c}_1}^{j+1}\right\|_0^2 + \left\|e_{\tilde{c}_2}^{j+1}\right\|_0^2\Big) + \frac{1}{7}\left\|\nabla e_{\tilde{c}_1}^{j+1}\right\|_0^2,\end{aligned} \tag{4.12}$$

and

$$\begin{aligned}\sum_{i=4}^{6}|E_i| &\le \Big(\left\|\tilde{c}_1^{j+1} - \tilde{c}_1^j\right\|_0 + \left\|\tilde{c}_1^j - R_h\tilde{c}_1^j\right\|_0 + \left\|e_{\tilde{c}_1}^j\right\|_0\Big)\left\|\nabla\psi^{j+1}\right\|_{0,\infty}\left\|\nabla e_{\tilde{c}_1}^{j+1}\right\|_0\\ &\le CC_0^2\left(\tau\int_{t_j}^{t_{j+1}}\|\tilde{c}_{1t}\|_0^2\,\mathrm{d}t + C_0^2h^4 + \left\|e_{\tilde{c}_1}^j\right\|_0^2\right) + \frac{1}{7}\left\|\nabla e_{\tilde{c}_1}^{j+1}\right\|_0^2.\end{aligned} \tag{4.13}$$

By use of the inductive hypothesis (3.4), Lemma **??** and Lemma 4.1, we obtain

$$|E_7| \le \left\|\tilde{c}_{1h}^j\right\|_{0,3}\left\|\nabla(\psi^{j+1} - R_h\psi^{j+1})\right\|_{0,6}\left\|\nabla e_{\tilde{c}_1}^{j+1}\right\|_0 \le CC_0^2C_1^2h^2 + \frac{1}{7}\left\|\nabla e_{\tilde{c}_1}^{j+1}\right\|_0^2, \tag{4.14}$$

$$|E_8| \le \left\|\tilde{c}_{1h}^j\right\|_{0,3}\left\|\nabla e_\psi^{j+1}\right\|_{0,6}\left\|\nabla e_{\tilde{c}_1}^{j+1}\right\|_0 \le CC_0^2C_1^2h^4 + CC_1^2\left(\left\|e_{\tilde{c}_1}^{j+1}\right\|_0^2 + \left\|e_{\tilde{c}_2}^{j+1}\right\|_0^2\right) + \frac{1}{7}\left\|\nabla e_{\tilde{c}_1}^{j+1}\right\|_0^2. \tag{4.15}$$

According to Lemma **??**, we infer

$$\begin{aligned}\sum_{i=9}^{11}|E_i| &\le C\left(\left\|u^{j+1}-u^j\right\|_0 + \left\|u^j - S_hu^j\right\|_0 + \left\|e_u^j\right\|_0\right)\left\|\tilde{c}_1^{j+1}\right\|_2\left\|\nabla e_{\tilde{c}_1}^{j+1}\right\|_0\\ &\le CC_0^2\left(\tau\int_{t_j}^{t_{j+1}}\|u_t\|_0^2\,\mathrm{d}t + C_0^2h^4 + \left\|e_u^j\right\|_0^2\right) + \frac{1}{7}\left\|\nabla e_{\tilde{c}_1}^{j+1}\right\|_0^2,\end{aligned} \tag{4.16}$$

and

$$\begin{aligned}|E_{12}| &\le \left|b_2(e_u^j;\, \tilde{c}_1^{j+1} - R_h\tilde{c}_1^{j+1},\, e_{\tilde{c}_1}^{j+1})\right| + \left|b_2(S_hu^j;\, \tilde{c}_1^{j+1} - R_h\tilde{c}_1^{j+1},\, e_{\tilde{c}_1}^{j+1})\right|\\ &\le \left\|\nabla e_u^j\right\|_0\left\|\nabla(\tilde{c}_1^{j+1} - R_h\tilde{c}_1^{j+1})\right\|_0\left\|\nabla e_{\tilde{c}_1}^{j+1}\right\|_0 + \|A_hS_hu^j\|_0\left\|\nabla e_{\tilde{c}_1}^{j+1}\right\|_0\left\|\tilde{c}_1^{j+1} - R_h\tilde{c}_1^{j+1}\right\|_0\\ &\le CC_0^2\left\|e_u^j\right\|_0^2 + CC_0^4h^4 + \frac{1}{7}\left\|\nabla e_{\tilde{c}_1}^{j+1}\right\|_0^2.\end{aligned} \tag{4.17}$$

Inserting (4.12)–(4.17) into (4.11), and summing up from $j = 0$ to $j = J - 1$, we get

$$\begin{aligned}\left\|e_{\tilde{c}_1}^J\right\|_0^2 + \tau\sum_{j=0}^{J-1}\left\|\nabla e_{\tilde{c}_1}^{j+1}\right\|_0^2 &\le CC_0^2(C_0^2 + C_1^2)(\tau^2 + h^2)\\ &\quad + C(C_0^2 + C_1^2)\,\tau\sum_{j=0}^{J-1}\left(\left\|e_{\tilde{c}_1}^{j+1}\right\|_0^2 + \left\|e_{\tilde{c}_2}^{j+1}\right\|_0^2 + \left\|e_u^{j+1}\right\|_0^2\right).\end{aligned} \tag{4.18}$$

Subsequently, the similar estimates for $e_{\tilde{c}_2}^{j+1}$ can also be obtained, which together with (4.18) lead to (4.10). □

Lemma 4.3. *Under the conditions of Lemma 4.1, we have*

$$\begin{aligned}\left\|e_u^J\right\|_0^2 + \tau\sum_{j=0}^{J-1}\left\|\nabla e_u^{j+1}\right\|_0^2 &\le CC_0^2(C_0^2 + C_1^2)(\tau^2 + h^4)\\ &\quad + C(C_0^2 + C_1^2)\,\tau\sum_{j=0}^{J-1}\left(\left\|e_{\tilde{c}_1}^{j+1}\right\|_0^2 + \left\|e_{\tilde{c}_2}^{j+1}\right\|_0^2 + \left\|e_u^{j+1}\right\|_0^2\right).\end{aligned} \tag{4.19}$$

Proof. Choose $v_h = e_u^{j+1} \in V_h^{\mathrm{div}}$ in (4.2), and note that

$$b_1(u_h^j;\, e_u^{j+1},\, e_u^{j+1}) = 0, \qquad \left(e_p^{j+1},\, \mathrm{div}\, e_u^{j+1}\right) = 0, \tag{4.20}$$

then we arrive at

$$\begin{aligned}&\frac{1}{2\tau}\left(\left\|e_u^{j+1}\right\|_0^2 + \left\|e_u^{j+1} - e_u^j\right\|_0^2 - \left\|e_u^j\right\|_0^2\right) + \left\|\nabla e_u^{j+1}\right\|_0^2\\ &= \left(u_t^{j+1} - D_\tau u^{j+1},\, e_u^{j+1}\right) + \left(D_\tau(u^{j+1} - S_hu^{j+1}),\, e_u^{j+1}\right) + b_1(u^{j+1} - u^j;\, u^{j+1},\, e_u^{j+1})\\ &\quad + b_1(u^j - S_hu^j;\, u^{j+1},\, e_u^{j+1}) + b_1(S_hu^j;\, u^{j+1} - S_hu^{j+1},\, e_u^{j+1}) - b_1(e_u^j;\, S_hu^{j+1},\, e_u^{j+1})\\ &\quad + \left(\tilde{c}_1^{j+1}\nabla\psi^{j+1} - \tilde{c}_{1h}^j\nabla\psi_h^{j+1},\, e_u^{j+1}\right) - \left(\tilde{c}_2^{j+1}\nabla\psi^{j+1} - \tilde{c}_{2h}^j\nabla\psi_h^{j+1},\, e_u^{j+1}\right) \;:=\; \sum_{i=1}^{8}G_i.\end{aligned} \tag{4.21}$$

By use of Taylor formula and Poincaré inequality, it is easy to check that

$$|G_1| \le C\tau\int_{t_j}^{t_{j+1}}\|u_{tt}\|_0^2\,\mathrm{d}t + \frac{1}{8}\left\|\nabla e_u^{j+1}\right\|_0^2. \tag{4.22}$$

Note that

$$\left\|D_\tau(u^{j+1} - S_hu^{j+1})\right\|_0^2 = \left\|\frac{(u^{j+1} - S_hu^{j+1}) - (u^j - S_hu^j)}{\tau}\right\|_0^2 = \left\|\frac{1}{\tau}\int_{t_j}^{t_{j+1}}(u - S_hu)_t\,\mathrm{d}t\right\|_0^2$$

$$\leq \left\| \frac{1}{\tau} \left(\int_{t_j}^{t_{j+1}} 1^2 \, \mathrm{d}t \right)^{1/2} \left(\int_{t_j}^{t_{j+1}} \|(u - S_h u)_t\|_0^2 \, \mathrm{d}t \right)^{1/2} \right\|_0^2 \leq \frac{1}{\tau} \int_{t_j}^{t_{j+1}} \|(u - S_h u)_t\|_0^2 \, \mathrm{d}t, \tag{4.23}$$

then using (**??**), we find

$$|G_2| \leq \frac{Ch^4}{\tau} \int_{t_j}^{t_{j+1}} \left(\|u_t\|_2 + \|p_t\|_1 \right) \mathrm{d}t + \frac{1}{8} \left\| \nabla e_u^{j+1} \right\|_0^2. \tag{4.24}$$

Due to Lemma **??** and Taylor formula, we deduce

$$|G_3| + |G_4| \leq C \left(\left\| u^{j+1} - u^j \right\|_0 + \left\| u^j - S_h u^j \right\|_0 \right) \left\| u^{j+1} \right\|_2 \left\| \nabla e_u^{j+1} \right\|_0 \leq C C_0^2 \tau \int_{t_j}^{t_{j+1}} \|u_t\|_0^2 \, \mathrm{d}t + C C_0^4 h^4 + \frac{1}{8} \left\| \nabla e_u^{j+1} \right\|_0^2, \tag{4.25}$$

and

$$|G_5| \leq C \|A_h S_h u^j\|_0 \left\| \nabla e_u^{j+1} \right\|_0 \left\| u^{j+1} - S_h u^{j+1} \right\|_0 \leq C C_0^4 h^4 + \frac{1}{8} \left\| \nabla e_u^{j+1} \right\|_0^2, \tag{4.26}$$

$$|G_6| \leq C \left\| e_u^j \right\|_0 \|A_h S_h u^{j+1}\|_0 \left\| \nabla e_u^{j+1} \right\|_0 \leq C C_0^2 \left\| e_u^j \right\|_0^2 + \frac{1}{8} \left\| \nabla e_u^{j+1} \right\|_0^2. \tag{4.27}$$

Next, G_7 can be split as

$$\begin{aligned} G_7 = {} & \left((\tilde{c}_1^{j+1} - \tilde{c}_1^j) \nabla \psi^{j+1}, \, e_u^{j+1} \right) + \left(\tilde{c}_1^j \nabla (\psi^{j+1} - R_h \psi^{j+1}), \, e_u^{j+1} \right) + \left((\tilde{c}_1^j - R_h \tilde{c}_1^j) \nabla R_h \psi^{j+1}, \, e_u^{j+1} \right) \\ & - \left(e_{\tilde{c}_1}^j \nabla R_h \psi^{j+1}, \, e_u^{j+1} \right) - \left(\tilde{c}_{1h}^j \nabla e_\psi^{j+1}, \, e_u^{j+1} \right) \; := \; \sum_{i=1}^{5} G_{7i}. \end{aligned} \tag{4.28}$$

It is not hard to find that

$$\begin{aligned} |G_{71}| + |G_{73}| + |G_{74}| + |G_{75}| \leq {} & \left\| \tilde{c}_1^{j+1} - \tilde{c}_1^j \right\|_0 \left\| \nabla \psi^{j+1} \right\|_{0,3} \left\| e_u^{j+1} \right\|_{0,6} + \left\| \tilde{c}_1^j - R_h \tilde{c}_1^j \right\|_0 \left\| \nabla R_h \psi^{j+1} \right\|_{0,3} \left\| e_u^{j+1} \right\|_{0,6} \\ & + \left\| e_{\tilde{c}_1}^j \right\|_0 \left\| \nabla R_h \psi^{j+1} \right\|_{0,3} \left\| e_u^{j+1} \right\|_{0,6} + \left\| \tilde{c}_{1h}^j \right\|_{0,3} \left\| \nabla e_\psi^{j+1} \right\|_0 \left\| e_u^{j+1} \right\|_{0,6} \\ \leq {} & C C_0^2 \tau \int_{t_j}^{t_{j+1}} \|\tilde{c}_{1t}\|_0^2 \, \mathrm{d}t + C C_0^2 (C_0^2 + C_1^2) h^4 + C C_0^2 \left\| e_{\tilde{c}_1}^j \right\|_0^2 \\ & + C C_1^2 \left(\left\| e_{\tilde{c}_1}^{j+1} \right\|_0^2 + \left\| e_{\tilde{c}_2}^{j+1} \right\|_0^2 \right) + \frac{1}{16} \left\| \nabla e_u^{j+1} \right\|_0^2. \end{aligned} \tag{4.29}$$

Using Green formula, we obtain

$$\begin{aligned} |G_{72}| = {} & \left| \left(\psi^{j+1} - R_h \psi^{j+1}, \, \nabla \tilde{c}_1^j \cdot e_u^{j+1} \right) + \left(\psi^{j+1} - R_h \psi^{j+1}, \, \tilde{c}_1^j \operatorname{div} e_u^{j+1} \right) \right| \\ \leq {} & \left\| \psi^{j+1} - R_h \psi^{j+1} \right\|_0 \left(\left\| \nabla \tilde{c}_1^j \right\|_{0,3} \left\| e_u^{j+1} \right\|_{0,6} + \left\| \tilde{c}_1^j \right\|_{0,\infty} \left\| \operatorname{div} e_u^{j+1} \right\|_0 \right) \leq C C_0^4 h^4 + \frac{1}{16} \left\| \nabla e_u^{j+1} \right\|_0^2. \end{aligned} \tag{4.30}$$

Combining (4.29) and (4.30), we have

$$\begin{aligned} |G_7| \leq {} & C C_0^2 \tau \int_{t_j}^{t_{j+1}} \|\tilde{c}_{1t}\|_0^2 \, \mathrm{d}t + C C_0^2 (C_0^2 + C_1^2) h^4 + C C_0^2 \left\| e_{\tilde{c}_1}^j \right\|_0^2 \\ & + C C_1^2 \left(\left\| e_{\tilde{c}_1}^{j+1} \right\|_0^2 + \left\| e_{\tilde{c}_2}^{j+1} \right\|_0^2 \right) + \frac{1}{8} \left\| \nabla e_u^{j+1} \right\|_0^2. \end{aligned} \tag{4.31}$$

In a similar way, G_8 can be estimated as

$$\begin{aligned} |G_8| \leq {} & C C_0^2 \tau \int_{t_j}^{t_{j+1}} \|\tilde{c}_{2t}\|_0^2 \, \mathrm{d}t + C C_0^2 (C_0^2 + C_1^2) h^4 + C C_0^2 \left\| e_{\tilde{c}_2}^j \right\|_0^2 \\ & + C C_1^2 \left(\left\| e_{\tilde{c}_1}^{j+1} \right\|_0^2 + \left\| e_{\tilde{c}_2}^{j+1} \right\|_0^2 \right) + \frac{1}{8} \left\| \nabla e_u^{j+1} \right\|_0^2. \end{aligned} \tag{4.32}$$

Therefore, inserting estimates (4.22), (4.24)–(4.27), (4.31), (4.32) into (4.21), we get

$$\frac{1}{2\tau} \left(\left\| e_u^{j+1} \right\|_0^2 - \left\| e_u^j \right\|_0^2 \right) + \frac{1}{8} \left\| \nabla e_u^{j+1} \right\|_0^2$$

$$\leq CC_0^2(C_0^2+C_1^2)h^4+\frac{Ch^4}{\tau}\int_{t_j}^{t_{j+1}}\left(\|u_t\|_2^2+\|p_t\|_1^2\right)\mathrm{d}t+CC_0^2\tau\int_{t_j}^{t_{j+1}}\left(\|u_{tt}\|_0^2+\|u_t\|_0^2+\|\tilde{c}_{1t}\|_0^2+\|\tilde{c}_{2t}\|_0^2\right)\mathrm{d}t$$
$$+CC_0^2\left\|e_u^j\right\|_0^2+C(C_0^2+C_1^2)\Big(\left\|e_{\tilde{c}_1}^j\right\|_0^2+\left\|e_{\tilde{c}_2}^j\right\|_0^2+\left\|e_{\tilde{c}_1}^{j+1}\right\|_0^2+\left\|e_{\tilde{c}_2}^{j+1}\right\|_0^2\Big). \tag{4.33}$$

Finally, summing (4.33) up from $j=0$ to $J-1$, we arrive at (4.19). □

Theorem 4.4. *Under the conditions of Lemma 4.1, if τ is small enough, we have*

$$\left\|e_{\tilde{c}_1}^J\right\|_0^2+\left\|e_{\tilde{c}_2}^J\right\|_0^2+\left\|e_u^J\right\|_0^2+\tau\sum_{j=0}^{J-1}\Big(\left\|\nabla e_{\tilde{c}_1}^{j+1}\right\|_0^2+\left\|\nabla e_{\tilde{c}_2}^{j+1}\right\|_0^2+\left\|\nabla e_u^{j+1}\right\|_0^2\Big)\leq C_2^2\big(\tau^2+h^2\big), \tag{4.34}$$

$$\left\|\nabla e_\psi^J\right\|_0+\left\|\Delta_h e_\psi^J\right\|_0\leq C_2(\tau+h), \tag{4.35}$$

where $C_2^2=CC_0^2(C_0^2+C_1^2)\exp\big(CC_0^2+CC_1^2\big)$.

Proof. Using Lemma 4.2 and Lemma 4.3, we get

$$\left\|e_{\tilde{c}_1}^J\right\|_0^2+\left\|e_{\tilde{c}_2}^J\right\|_0^2+\left\|e_u^J\right\|_0^2+\tau\sum_{j=0}^{J-1}\Big(\left\|\nabla e_{\tilde{c}_1}^{j+1}\right\|_0^2+\left\|\nabla e_{\tilde{c}_2}^{j+1}\right\|_0^2+\left\|\nabla e_u^{j+1}\right\|_0^2\Big)$$
$$\leq CC_0^2(C_0^2+C_1^2)(\tau^2+h^2)+C(C_0^2+C_1^2)\,\tau\sum_{j=0}^{J-1}\Big(\left\|e_{\tilde{c}_1}^{j+1}\right\|_0^2+\left\|e_{\tilde{c}_2}^{j+1}\right\|_0^2+\left\|e_u^{j+1}\right\|_0^2\Big). \tag{4.36}$$

By the Gronwall lemma, taking $\tau<\tau_2:=\dfrac{1}{C(C_0^2+C_1^2)}$, we deduce (4.34). Then using Lemma 4.1, we obtain (4.35). □

The error estimates in Lemmas 4.1–4.3 and Theorem 4.4 were derived under the inductive hypothesis (3.4). Now, we check it.

When $j=0$, we deduce

$$\left\|\tilde{c}_{1h}^0\right\|_{0,3}+\left\|\tilde{c}_{2h}^0\right\|_{0,3}=\|R_h\tilde{c}_{10}\|_{0,3}+\|R_h\tilde{c}_{20}\|_{0,3}<1+C_0=:C_1. \tag{4.37}$$

Assume that

$$\left\|\tilde{c}_{1h}^m\right\|_{0,3}+\left\|\tilde{c}_{2h}^m\right\|_{0,3}\leq C_1 \tag{4.38}$$

is true for $m\leq j$. Next, we derive (3.4) for $m=j+1$. To this aim, we consider two cases. If $\tau<h$, we have from the inverse inequality and Theorem 4.4 that

$$\left\|e_{\tilde{c}_1}^{j+1}\right\|_{0,3}+\left\|e_{\tilde{c}_2}^{j+1}\right\|_{0,3}\leq Ch^{-d/6}\Big(\left\|e_{\tilde{c}_1}^{j+1}\right\|_0+\left\|e_{\tilde{c}_2}^{j+1}\right\|_0\Big)\leq CC_2h^{1/2}. \tag{4.39}$$

If $h<\tau$, by Theorem 4.4, we deduce

$$\left\|e_{\tilde{c}_1}^{j+1}\right\|_{0,3}+\left\|e_{\tilde{c}_2}^{j+1}\right\|_{0,3}\leq C\Big(\left\|\nabla e_{\tilde{c}_1}^{j+1}\right\|_0+\left\|\nabla e_{\tilde{c}_2}^{j+1}\right\|_0\Big)\leq CC_2\tau^{1/2}. \tag{4.40}$$

In any case, if $\tau<\tau_3:=\dfrac{1}{C^2C_2^2}$ and $h<h_1:=\dfrac{1}{C^2C_2^2}$, we conclude that

$$\left\|e_{\tilde{c}_1}^{j+1}\right\|_{0,3}+\left\|e_{\tilde{c}_2}^{j+1}\right\|_{0,3}<1. \tag{4.41}$$

Thus, we have

$$\left\|\tilde{c}_{1h}^{j+1}\right\|_{0,3}+\left\|\tilde{c}_{2h}^{j+1}\right\|_{0,3}\leq\left\|e_{\tilde{c}_1}^{j+1}\right\|_{0,3}+\left\|e_{\tilde{c}_2}^{j+1}\right\|_{0,3}+\left\|R_h\tilde{c}_1^{j+1}\right\|_{0,3}+\left\|R_h\tilde{c}_2^{j+1}\right\|_{0,3}\leq 1+C_0=C_1. \tag{4.42}$$

The check of the inductive hypothesis (3.4) is completed.

Remark 4.5. In the using of the mathematical inductive method, it is crucial to bound $\|\tilde{c}_{1h}^{j+1}\|_{0,3}$ and $\|\tilde{c}_{2h}^{j+1}\|_{0,3}$ to an appointed constant, otherwise the error bounds in Lemmas 4.1–4.3 and Theorem 4.4 may become larger and larger.

Remark 4.6. Theorem 4.4 shows that the error estimates of the $\ell^\infty(L^2)$ norm are suboptimal, which is caused by the fact that the estimate of the term E_7 in Lemma 4.2 cannot achieve optimal.

Finally, we derive the unconditional stability of solutions.

Lemma 4.7. *Under the conditions of Lemma 4.1, if τ and h are small enough, we have*

$$\left\|\nabla\psi_h^{j+1}\right\|_0 + \left\|\Delta_h\psi_h^{j+1}\right\|_0 \le C_3, \tag{4.43}$$

$$\left\|\nabla\tilde{c}_{1h}^{j+1}\right\|_0 + \left\|\nabla\tilde{c}_{2h}^{j+1}\right\|_0 + \left\|\nabla u_h^{j+1}\right\|_0 \le C_3, \tag{4.44}$$

$$\tau\sum_{j=0}^{J-1}\Big(\left\|\Delta_h\tilde{c}_{1h}^{j+1}\right\|_0^2 + \left\|\Delta_h\tilde{c}_{2h}^{j+1}\right\|_0^2 + \left\|A_h u_h^{j+1}\right\|_0^2\Big) \le C_4^2, \tag{4.45}$$

where $C_3 = CC_2 + 2C_0$, $C_4^2 = CC_3^6 + CC_0^2$.

Proof. First of all, from the definition of the Ritz projection (**??**), we have

$$\big(\nabla R_h\psi^{j+1},\, \nabla\chi_h\big) = \big(\nabla\psi^{j+1},\, \nabla\chi_h\big), \quad \forall\,\chi_h \in X_h^0. \tag{4.46}$$

Taking $\chi_h = R_h\psi^{j+1}$ and $\chi_h = \Delta_h R_h\psi^{j+1}$ in (4.46), respectively, and using (**??**), we infer

$$\left\|\nabla R_h\psi^{j+1}\right\|_0^2 = \big(\nabla\psi^{j+1},\, \nabla R_h\psi^{j+1}\big) \le \left\|\nabla\psi^{j+1}\right\|_0\left\|\nabla R_h\psi^{j+1}\right\|_0, \tag{4.47}$$

$$\begin{aligned}\left\|\Delta_h R_h\psi^{j+1}\right\|_0^2 &= \big(\nabla R_h\psi^{j+1},\, \nabla\Delta_h R_h\psi^{j+1}\big) = \big(\nabla\psi^{j+1},\, \nabla\Delta_h R_h\psi^{j+1}\big)\\ &= -\big(\Delta\psi^{j+1},\, \Delta_h R_h\psi^{j+1}\big) \le \left\|\Delta\psi^{j+1}\right\|_0\left\|\Delta_h R_h\psi^{j+1}\right\|_0,\end{aligned} \tag{4.48}$$

which implies

$$\left\|\nabla R_h\psi^{j+1}\right\|_0 \le \left\|\nabla\psi^{j+1}\right\|_0, \qquad \left\|\Delta_h R_h\psi^{j+1}\right\|_0 \le \left\|\Delta\psi^{j+1}\right\|_0. \tag{4.49}$$

Then using (4.35) in Theorem 4.4, we deduce

$$\begin{aligned}\left\|\nabla\psi_h^{j+1}\right\|_0 + \left\|\Delta_h\psi_h^{j+1}\right\|_0 &= \left\|\nabla e_\psi^{j+1}\right\|_0 + \left\|\Delta_h e_\psi^{j+1}\right\|_0 + \left\|\nabla R_h\psi^{j+1}\right\|_0 + \left\|\Delta_h R_h\psi^{j+1}\right\|_0\\ &\le C_2(\tau+h) + 2\left\|\psi^{j+1}\right\|_2 \le CC_2 + 2C_0 =:\, C_3.\end{aligned} \tag{4.50}$$

Next, if $\tau < h$, from the inverse inequality and the $\ell^\infty(L^2)$ norm estimate in (4.34), we infer

$$\left\|\nabla e_{\tilde{c}_1}^{j+1}\right\|_0 \le Ch^{-1}\left\|e_{\tilde{c}_1}^{j+1}\right\|_0 \le CC_2h^{-1}(\tau+h) \le CC_2. \tag{4.51}$$

If $h < \tau$, by use of the $\ell^2(H^1)$ norm estimate in (4.34), we deduce

$$\left\|\nabla e_{\tilde{c}_1}^{j+1}\right\|_0 \le C_2\tau^{-1/2}(\tau+h) \le CC_2\tau^{1/2}. \tag{4.52}$$

In any case, we obtain

$$\left\|\nabla\tilde{c}_{1h}^{j+1}\right\|_0 \le \left\|\nabla e_{\tilde{c}_1}^{j+1}\right\|_0 + \left\|\nabla R_h\tilde{c}_1^{j+1}\right\|_0 \le CC_2 + C_0 =:\, C_3. \tag{4.53}$$

Similarly, we can derive the bounds of $\ell^\infty(H^1)$ norm for $\tilde{c}_{2h}^{j+1}$ and u_h^{j+1}, which implies (4.44).

Finally, taking $\eta_{1h} = 2\Delta_h\tilde{c}_{1h}^{j+1}\tau$ in the third equation of (3.2), and using (**??**), (**??**), (**??**), we have

$$\begin{aligned}&\left\|\nabla\tilde{c}_{1h}^{j+1}\right\|_0^2 - \left\|\nabla\tilde{c}_{1h}^{j}\right\|_0^2 + \left\|\nabla(\tilde{c}_{1h}^{j+1} - \tilde{c}_{1h}^{j})\right\|_0^2 + 2\tau\left\|\Delta_h\tilde{c}_{1h}^{j+1}\right\|_0^2\\ &\le 2\tau\big(\tilde{c}_{1h}^{j}\nabla\psi_h^{j+1},\, \nabla\Delta_h\tilde{c}_{1h}^{j+1}\big) + 2\tau m_0\big(\nabla\psi_h^{j+1},\, \nabla\Delta_h\tilde{c}_{1h}^{j+1}\big) + 2\tau b_2\big(u_h^j;\, \tilde{c}_{1h}^{j+1},\, \Delta_h\tilde{c}_{1h}^{j+1}\big)\\ &\le C\tau\Big(\left\|\nabla\tilde{c}_{1h}^{j}\right\|_0^{1/2}\left\|\Delta_h\tilde{c}_{1h}^{j}\right\|_0^{1/2} + m_0\Big)\left\|\Delta_h\psi_h^{j+1}\right\|_0\left\|\Delta_h\tilde{c}_{1h}^{j+1}\right\|_0\\ &\quad + C\tau\left\|\nabla u_h^j\right\|_0\left\|\nabla\tilde{c}_{1h}^{j+1}\right\|_0^{1/2}\left\|\Delta_h\tilde{c}_{1h}^{j+1}\right\|_0^{3/2}\\ &\le CC_3^6\tau + \tau\left\|\Delta_h\tilde{c}_{1h}^{j+1}\right\|_0^2.\end{aligned} \tag{4.54}$$

Summing (4.54) up from $j = 0$ to $J-1$, we get

$$\tau\sum_{j=0}^{J-1}\left\|\Delta_h\tilde{c}_{1h}^{j+1}\right\|_0^2 \le CC_3^6 + C\left\|\nabla\tilde{c}_{1h}^{0}\right\|_0^2 \le CC_3^6 + CC_0^2 =:\, C_4^2. \tag{4.55}$$

In an analogous procedure, we can derive the rest part of (4.45). The proof is completed. □

4.2. The $\ell^\infty(H^1)$-norm error estimate

In this section, we derive the $\ell^\infty(H^1)$-norm error estimates of all the variables, which are shown in following Theorems 4.8–4.10.

Theorem 4.8. *Under the conditions of Lemma 4.1, we have*

$$\tau\sum_{j=0}^{J-1}\left\|D_\tau e_u^{j+1}\right\|_0^2+\left\|\nabla e_u^J\right\|_0^2\le C_5^2\left(\tau^2+h^2\right), \tag{4.56}$$

where $C_5^2=C(C_0^2+C_1^2)C_2^2\exp\left(CC_0^2+CC_4^2\right)$.

Proof. Putting $v_h=D_\tau e_u^{j+1}$ in (4.2) results in

$$\begin{aligned}
&\left\|D_\tau e_u^{j+1}\right\|_0^2+\frac{1}{2\tau}\left(\left\|\nabla e_u^{j+1}\right\|_0^2-\left\|\nabla e_u^j\right\|_0^2\right)\\
&\le\left(u_t^{j+1}-D_\tau u^{j+1},\,D_\tau e_u^{j+1}\right)+\left(D_\tau(u^{j+1}-S_hu^{j+1}),\,D_\tau e_u^{j+1}\right)+b_1(u^{j+1}-u^j;\,u^{j+1},\,D_\tau e_u^{j+1})\\
&\quad+b_1(u^j-S_hu^j;\,u^{j+1},\,D_\tau e_u^{j+1})+b_1(S_hu^j;\,u^{j+1}-S_hu^{j+1},\,D_\tau e_u^{j+1})-b_1(e_u^j;\,S_hu^{j+1},\,D_\tau e_u^{j+1})\\
&\quad-b_1(u_h^j;\,e_u^{j+1},\,D_\tau e_u^{j+1})+\left(\tilde c_1^{j+1}\nabla\psi^{j+1}-\tilde c_{1h}^j\nabla\psi_h^{j+1},\,D_\tau e_u^{j+1}\right)-\left(\tilde c_2^{j+1}\nabla\psi^{j+1}-\tilde c_{2h}^j\nabla\psi_h^{j+1},\,D_\tau e_u^{j+1}\right)\ :=\ \sum_{i=1}^9 L_i.
\end{aligned} \tag{4.57}$$

Now, we estimate the above nine terms. It is not difficult to see that

$$|L_1|+|L_2|\le C\tau\int_{t_j}^{t_{j+1}}\|u_{tt}\|_0^2\,\mathrm{d}t+\frac{Ch^4}{\tau}\int_{t_j}^{t_{j+1}}\left(\|u_t\|_2^2+\|p_t\|_1^2\right)\mathrm{d}t+\frac{1}{8}\left\|D_\tau e_u^{j+1}\right\|_0^2. \tag{4.58}$$

In view of Lemma **??**, we infer

$$\begin{aligned}
|L_3|+|L_4|&\le C\left(\left\|\nabla(u^{j+1}-u^j)\right\|_0+\left\|\nabla(u^j-S_hu^j)\right\|_0\right)\left\|u^{j+1}\right\|_2\left\|D_\tau e_u^{j+1}\right\|_0\\
&\le CC_0^2\tau\int_{t_j}^{t_{j+1}}\|\nabla u_t\|_0^2\,\mathrm{d}t+CC_0^4h^2+\frac{1}{8}\left\|D_\tau e_u^{j+1}\right\|_0^2,
\end{aligned} \tag{4.59}$$

and

$$|L_5|\le\|A_hS_hu^j\|_0\left\|\nabla(u^{j+1}-S_hu^{j+1})\right\|_0\left\|D_\tau e_u^{j+1}\right\|_0\le CC_0^4h^2+\frac{1}{8}\left\|D_\tau e_u^{j+1}\right\|_0^2, \tag{4.60}$$

$$|L_6|\le\left\|\nabla e_u^j\right\|_0\left\|A_hS_hu^{j+1}\right\|_0\left\|D_\tau e_u^{j+1}\right\|_0\le CC_0^2\left\|\nabla e_u^j\right\|_0^2+\frac{1}{8}\left\|D_\tau e_u^{j+1}\right\|_0^2, \tag{4.61}$$

$$\begin{aligned}
|L_7|&=\left|b_1(u_h^j;\,e_u^{j+1}-e_u^j,\,D_\tau e_u^{j+1})+b_1(u_h^j;\,e_u^j,\,D_\tau e_u^{j+1})\right|\\
&\le 0+C\left\|A_hu_h^j\right\|_0\left\|\nabla e_u^j\right\|_0\left\|D_\tau e_u^{j+1}\right\|_0\\
&\le C\left\|A_hu_h^j\right\|_0^2\left\|\nabla e_u^j\right\|_0^2+\frac{1}{8}\left\|D_\tau e_u^{j+1}\right\|_0^2.
\end{aligned} \tag{4.62}$$

The term L_8 can be split as

$$\begin{aligned}
L_8&=\left((\tilde c_1^{j+1}-\tilde c_1^j)\nabla\psi^{j+1},\,D_\tau e_u^{j+1}\right)+\left(\tilde c_1^j\nabla(\psi^{j+1}-R_h\psi^{j+1}),\,D_\tau e_u^{j+1}\right)\\
&\quad+\left((\tilde c_1^j-R_h\tilde c_1^j)\nabla R_h\psi^{j+1},\,D_\tau e_u^{j+1}\right)-\left(e_{\tilde c_1}^j\nabla R_h\psi^{j+1},\,D_\tau e_u^{j+1}\right)-\left(\tilde c_{1h}^j\nabla e_\psi^{j+1},\,D_\tau e_u^{j+1}\right).
\end{aligned} \tag{4.63}$$

Then we deduce

$$\begin{aligned}
L_8&\le\left\|\tilde c_1^{j+1}-\tilde c_1^j\right\|_0\left\|\nabla\psi^{j+1}\right\|_{0,\infty}\left\|D_\tau e_u^{j+1}\right\|_0+\left\|\tilde c_1^j\right\|_{0,\infty}\left\|\nabla(\psi^{j+1}-R_h\psi^{j+1})\right\|_0\left\|D_\tau e_u^{j+1}\right\|_0\\
&\quad+\left(\left\|\tilde c_1^j-R_h\tilde c_1^j\right\|_0+\left\|e_{\tilde c_1}^j\right\|_0\right)\left\|\nabla R_h\psi^{j+1}\right\|_{0,\infty}\left\|D_\tau e_u^{j+1}\right\|_0+\left\|\tilde c_{1h}^j\right\|_{0,3}\left\|\nabla e_\psi^{j+1}\right\|_{0,6}\left\|D_\tau e_u^{j+1}\right\|_0\\
&\le CC_0^2\tau\int_{t_j}^{t_{j+1}}\|\tilde c_{1t}\|_0^2\,\mathrm{d}t+C(C_0^2+C_1^2)C_2^2(\tau^2+h^2)+\frac{1}{8}\left\|D_\tau e_u^{j+1}\right\|_0^2.
\end{aligned} \tag{4.64}$$

Similarly, we get

$$L_9\le CC_0^2\tau\int_{t_j}^{t_{j+1}}\|\tilde c_{2t}\|_0^2\,\mathrm{d}t+C(C_0^2+C_1^2)C_2^2(\tau^2+h^2)+\frac{1}{8}\left\|D_\tau e_u^{j+1}\right\|_0^2. \tag{4.65}$$

Inserting estimates (4.58)–(4.65) into (4.57), and then summing up from $j = 0$ to $J - 1$, we have

$$\tau \sum_{j=0}^{J-1} \left\| D_\tau e_u^{j+1} \right\|_0^2 + \left\| \nabla e_u^J \right\|_0^2 \le C(C_0^2 + C_1^2)C_2^2(\tau^2 + h^2) + C\tau \sum_{j=0}^{J-1} \left(C_0^2 + \left\| A_h u_h^j \right\|_0^2 \right) \left\| \nabla e_u^j \right\|_0^2. \tag{4.66}$$

Finally, by the Gronwall lemma and (4.45), we conclude (4.56). □

Theorem 4.9. *Under the conditions of Lemma 4.1, we have*

$$\left\| \nabla e_{\tilde{c}_1}^J \right\|_0^2 + \left\| \nabla e_{\tilde{c}_2}^J \right\|_0^2 + \tau \sum_{j=0}^{J-1} \left(\left\| D_\tau e_{\tilde{c}_1}^{j+1} \right\|_0^2 + \left\| D_\tau e_{\tilde{c}_2}^{j+1} \right\|_0^2 \right) \le C_6^2 \left(\tau^2 + h^2 \right), \tag{4.67}$$

where $C_6^2 = C(C_2^2 C_4^2 + C_0^2 C_5^2) \exp\left(C C_0^2 + C C_4^2 \right)$.

Proof. Set $\eta_{1h} = D_\tau e_{\tilde{c}_1}^{j+1}$ in (4.4), and we obtain

$$\begin{aligned}
&\left\| D_\tau e_{\tilde{c}_1}^{j+1} \right\|_0^2 + \frac{1}{2\tau} \left(\left\| \nabla e_{\tilde{c}_1}^{j+1} \right\|_0^2 - \left\| \nabla e_{\tilde{c}_1}^j \right\|_0^2 \right) \\
&= \left(\tilde{c}_{1t}^{j+1} - D_\tau \tilde{c}_1^{j+1},\ D_\tau e_{\tilde{c}_1}^{j+1} \right) + \left(D_\tau \tilde{c}_1^{j+1} - D_\tau R_h \tilde{c}_1^{j+1},\ D_\tau e_{\tilde{c}_1}^{j+1} \right) - m_0 \left(\nabla e_\psi^{j+1},\ \nabla D_\tau e_{\tilde{c}_1}^{j+1} \right) \\
&\quad + \left((\tilde{c}_1^{j+1} - \tilde{c}_1^j) \nabla \psi^{j+1},\ \nabla D_\tau e_{\tilde{c}_1}^{j+1} \right) + \left((\tilde{c}_1^j - R_h \tilde{c}_1^j) \nabla \psi^{j+1},\ \nabla D_\tau e_{\tilde{c}_1}^{j+1} \right) - \left(e_{\tilde{c}_1}^j \nabla \psi^{j+1},\ \nabla D_\tau e_{\tilde{c}_1}^{j+1} \right) \\
&\quad + \left(\tilde{c}_{1h}^j \nabla (\psi^{j+1} - R_h \psi^{j+1}),\ \nabla D_\tau e_{\tilde{c}_1}^{j+1} \right) - \left(\tilde{c}_{1h}^j \nabla e_\psi^{j+1},\ \nabla D_\tau e_{\tilde{c}_1}^{j+1} \right) + b_2(u^{j+1} - u^j;\ \tilde{c}_1^{j+1},\ D_\tau e_{\tilde{c}_1}^{j+1}) \\
&\quad + b_2(u^j - S_h u^j;\ \tilde{c}_1^{j+1},\ D_\tau e_{\tilde{c}_1}^{j+1}) - b_2(e_u^j;\ \tilde{c}_1^{j+1},\ D_\tau e_{\tilde{c}_1}^{j+1}) + b_2(u_h^j;\ \tilde{c}_1^{j+1} - R_h \tilde{c}_1^{j+1},\ D_\tau e_{\tilde{c}_1}^{j+1}) \\
&\quad - b_2(u_h^j;\ e_{\tilde{c}_1}^{j+1},\ D_\tau e_{\tilde{c}_1}^{j+1}) \ := \ \sum_{i=1}^{13} M_i.
\end{aligned} \tag{4.68}$$

It is not hard to find that

$$|M_1| + |M_2| \le C\tau \int_{t_j}^{t_{j+1}} \|\tilde{c}_{1tt}\|_0^2 \, \mathrm{d}t + \frac{Ch^4}{\tau} \int_{t_j}^{t_{j+1}} \|\tilde{c}_{1t}\|_2^2 \, \mathrm{d}t + \frac{1}{10} \left\| D_\tau e_{\tilde{c}_1}^{j+1} \right\|_0^2, \tag{4.69}$$

and

$$|M_5| \le C \left\| \tilde{c}_1^j - R_h \tilde{c}_1^j \right\|_0 \left\| \nabla \psi^{j+1} \right\|_{0,\infty} \left\| \nabla D_\tau e_{\tilde{c}_1}^{j+1} \right\|_0 \le C C_0^4 h^2 + \frac{1}{10} \left\| D_\tau e_{\tilde{c}_1}^{j+1} \right\|_0^2. \tag{4.70}$$

According to (**??**), we find

$$|M_3| \le m_0 \left\| \Delta_h e_\psi^{j+1} \right\|_0 \left\| D_\tau e_{\tilde{c}_1}^{j+1} \right\|_0 \le C C_2^2 (\tau^2 + h^2) + \frac{1}{10} \left\| D_\tau e_{\tilde{c}_1}^{j+1} \right\|_0^2. \tag{4.71}$$

By use of Green formula, we get

$$\begin{aligned}
|M_4| &= \left| \left(\nabla (\tilde{c}_1^{j+1} - \tilde{c}_1^j) \cdot \nabla \psi^{j+1},\ D_\tau e_{\tilde{c}_1}^{j+1} \right) + \left((\tilde{c}_1^{j+1} - \tilde{c}_1^j) \Delta \psi^{j+1},\ D_\tau e_{\tilde{c}_1}^{j+1} \right) \right| \\
&\le C C_0^2 \tau \int_{t_j}^{t_{j+1}} \|\nabla \tilde{c}_{1t}\|_0^2 \, \mathrm{d}t + \frac{1}{10} \left\| D_\tau e_{\tilde{c}_1}^{j+1} \right\|_0^2,
\end{aligned} \tag{4.72}$$

and

$$|M_6| = \left| \left(\nabla e_{\tilde{c}_1}^j \cdot \nabla \psi^{j+1},\ D_\tau e_{\tilde{c}_1}^{j+1} \right) + \left(e_{\tilde{c}_1}^j \Delta \psi^{j+1},\ D_\tau e_{\tilde{c}_1}^{j+1} \right) \right| \le C C_0^2 \left\| \nabla e_{\tilde{c}_1}^j \right\|_0^2 + \frac{1}{10} \left\| D_\tau e_{\tilde{c}_1}^{j+1} \right\|_0^2. \tag{4.73}$$

Thanks to Lemma **??**, we infer

$$|M_8| \le C \left\| \Delta_h \tilde{c}_{1h}^j \right\|_0 \left\| \Delta_h e_\psi^{j+1} \right\|_0 \left\| D_\tau e_{\tilde{c}_1}^{j+1} \right\|_0 \le C C_2^2 (\tau^2 + h^2) \left\| \Delta_h \tilde{c}_{1h}^j \right\|_0^2 + \frac{1}{10} \left\| D_\tau e_{\tilde{c}_1}^{j+1} \right\|_0^2. \tag{4.74}$$

In view of Lemma **??**, we deduce

$$\sum_{i=9}^{11} |M_i| \le \left(\left\| \nabla (u^{j+1} - u^j) \right\|_0 + \left\| \nabla (u^j - S_h u^j) \right\|_0 + \left\| \nabla e_u^j \right\|_0 \right) \left\| \tilde{c}_1^{j+1} \right\|_2 \left\| D_\tau e_{\tilde{c}_1}^{j+1} \right\|_0$$

$$\leq CC_0^2C_5^2(\tau^2+h^2)+CC_0^2\tau^2\int_{t_j}^{t_{j+1}}\|\nabla u_t\|_0^2\,\mathrm{d}t+\frac{1}{10}\left\|D_\tau e_{\tilde{c}_1}^{j+1}\right\|_0^2, \tag{4.75}$$

$$\begin{aligned}\sum_{i=12}^{13}|M_i| &\leq C\left\|A_hu_h^j\right\|_0\left(\left\|\nabla(\tilde{c}_1^{j+1}-R_h\tilde{c}_1^{j+1})\right\|_0+\left\|\nabla e_{\tilde{c}_1}^j\right\|_0\right)\left\|D_\tau e_{\tilde{c}_1}^{j+1}\right\|_0\\ &\leq CC_0^2\left\|A_hu_h^j\right\|_0^2h^2+C\left\|A_hu_h^j\right\|_0^2\left\|\nabla e_{\tilde{c}_1}^j\right\|_0^2+\frac{1}{10}\left\|D_\tau e_{\tilde{c}_1}^{j+1}\right\|_0^2.\end{aligned} \tag{4.76}$$

The term M_7 can be split as

$$\begin{aligned}M_7=&\left(\left(e_{\tilde{c}_1}^j+(R_h\tilde{c}_1^j-\tilde{c}_1^j)+(\tilde{c}_1^j-\tilde{c}_1^{j+1})\right)\nabla(\psi^{j+1}-R_h\psi^{j+1}),\ \nabla D_\tau e_{\tilde{c}_1}^{j+1}\right)\\ &+\left(\tilde{c}_1^{j+1}\nabla(\psi^{j+1}-R_h\psi^{j+1}),\ \nabla D_\tau e_{\tilde{c}_1}^{j+1}\right),\end{aligned} \tag{4.77}$$

and note that

$$\begin{aligned}\left(\tilde{c}_1^{j+1}\nabla(\psi^{j+1}-R_h\psi^{j+1}),\ \nabla D_\tau e_{\tilde{c}_1}^{j+1}\right)=&\ D_\tau\left(\tilde{c}_1^{j+1}\nabla(\psi^{j+1}-R_h\psi^{j+1}),\ \nabla e_{\tilde{c}_1}^{j+1}\right)\\ &-\left(\tilde{c}_1^{j+1}\nabla D_\tau(\psi^{j+1}-R_h\psi^{j+1}),\ \nabla e_{\tilde{c}_1}^j\right)\\ &-\left(D_\tau\tilde{c}_1^{j+1}\nabla(\psi^j-R_h\psi^j),\ \nabla e_{\tilde{c}_1}^{j+1}\right),\end{aligned} \tag{4.78}$$

thus, we obtain

$$\begin{aligned}|M_7|\leq&\left(\left\|e_{\tilde{c}_1}^j\right\|_{0,6}+\left\|R_h\tilde{c}_1^j-\tilde{c}_1^j\right\|_{0,6}+\left\|\tilde{c}_1^{j+1}-\tilde{c}_1^j\right\|_{0,6}\right)\left\|\nabla(\psi^{j+1}-R_h\psi^{j+1})\right\|_{0,3}\left\|\nabla D_\tau e_{\tilde{c}_1}^{j+1}\right\|_0\\ &+\left\|\tilde{c}_1^{j+1}\right\|_{0,\infty}\left\|\nabla(D_\tau\psi^{j+1}-R_hD_\tau\psi^{j+1})\right\|_0\left\|\nabla e_{\tilde{c}_1}^j\right\|_0\\ &+\left\|D_\tau\tilde{c}_1^{j+1}\right\|_{0,\infty}\left\|\nabla(\psi^j-R_h\psi^j)\right\|_0\left\|\nabla e_{\tilde{c}_1}^{j+1}\right\|_0+D_\tau\left(\tilde{c}_1^{j+1}\nabla(\psi^{j+1}-R_h\psi^{j+1}),\ \nabla e_{\tilde{c}_1}^{j+1}\right)\\ \leq&\ CC_0^2\left(\left\|\nabla e_{\tilde{c}_1}^j\right\|_0^2+C_0^2h^2+\tau\int_{t_j}^{t_{j+1}}\|\nabla\tilde{c}_{1t}\|_0^2\,\mathrm{d}t\right)+\frac{1}{10}\left\|D_\tau e_{\tilde{c}_1}\right\|_0^2\\ &+\frac{CC_0^2h^2}{\tau}\int_{t_j}^{t_{j+1}}\left(\|\psi_t\|_2^2+\|\tilde{c}_{1t}\|_2^2\right)\mathrm{d}t+C\left\|\nabla e_{\tilde{c}_1}^j\right\|_0^2+D_\tau\left(\tilde{c}_1^{j+1}\nabla(\psi^{j+1}-R_h\psi^{j+1}),\ \nabla e_{\tilde{c}_1}^{j+1}\right).\end{aligned} \tag{4.79}$$

Inserting (4.69)–(4.79) into (4.68), then summing up from $j=0$ to $J-1$, we deduce

$$\begin{aligned}&\tau\sum_{j=0}^{J-1}\left\|D_\tau e_{\tilde{c}_1}^{j+1}\right\|_0^2+\left\|\nabla e_{\tilde{c}_1}^J\right\|_0^2\\ &\leq C(C_2^2C_4^2+C_0^2C_5^2)(\tau^2+h^2)+C\tau\sum_{j=0}^{J-1}\left(C_0^2+\left\|A_hu_h^j\right\|_0^2\right)\left\|\nabla e_{\tilde{c}_1}^j\right\|_0^2+\left(\tilde{c}_1^J\nabla(\psi^J-R_h\psi^J),\ \nabla e_{\tilde{c}_1}^J\right).\end{aligned} \tag{4.80}$$

In addition, observing that

$$\left|\left(\tilde{c}_1^J\nabla(\psi^J-R_h\psi^J),\ \nabla e_{\tilde{c}_1}^J\right)\right|\leq\left\|\tilde{c}_1^J\right\|_{0,\infty}\left\|\nabla(\psi^J-R_h\psi^J)\right\|_0\left\|\nabla e_{\tilde{c}_1}^J\right\|_0\leq CC_0^4h^2+\frac{1}{2}\left\|\nabla e_{\tilde{c}_1}^J\right\|_0^2, \tag{4.81}$$

then substituting (4.81) into (4.80), and using the Gronwall lemma, we have

$$\tau\sum_{j=0}^{J-1}\left\|D_\tau e_{\tilde{c}_1}^{j+1}\right\|_0^2+\left\|\nabla e_{\tilde{c}_1}^J\right\|_0^2\leq C(C_2^2C_4^2+C_0^2C_5^2)\exp\left(CC_0^2+CC_4^2\right)(\tau^2+h^2). \tag{4.82}$$

A similar estimate can be obtained for $e_{\tilde{c}_2}^{j+1}$. The proof is completed. □

Finally, based on Theorems 4.4–4.9, and using the triangle inequality, we have

Theorem 4.10. *Under the conditions of Lemma 4.1, we have*

$$\left\|c_1^{j+1}-c_{1h}^{j+1}\right\|_0+\left\|c_2^{j+1}-c_{2h}^{j+1}\right\|_0+\left\|u^{j+1}-u_h^{j+1}\right\|_0+\left\|\nabla(\psi^{j+1}-\psi_h^{j+1})\right\|_0\leq C_2(\tau+h), \tag{4.83}$$

$$\left\|\nabla(u^{j+1}-u_h^{j+1})\right\|_0\leq C_5(\tau+h), \tag{4.84}$$

$$\left\|\nabla(c_1^{j+1}-c_{1h}^{j+1})\right\|_0+\left\|\nabla(c_2^{j+1}-c_{2h}^{j+1})\right\|_0\leq C_6(\tau+h). \tag{4.85}$$

4.3. Improved error estimate for the $\ell^\infty(L^2)$-norm

In this section, let Ω be a rectangular domain with the boundary $\partial\Omega$ parallel to x_1-axis or x_2-axis. $\mathcal{T}_h$ is a family of axis-parallel rectangular meshes of Ω, and K is a rectangular element. We use the bilinear element to solve c_1, c_2, ψ, and Q_1-Q_0 element for u and p [30].

Then, we have the following improved estimate.

Lemma 4.11. *For all $\psi \in H^3(\Omega)$, $\varphi_h \in X_h^0$, there holds*

$$\big(\nabla(\psi - R_h\psi),\, \nabla\varphi_h\big) \le Ch^2\|\psi\|_3\|\nabla\varphi_h\|_0. \tag{4.86}$$

Proof. Let I_h be the interpolation operator on X_h^0, then we obtain from [36] that

$$\big(\nabla(\psi - I_h\psi),\, \nabla\varphi_h\big) \le Ch^2\|\psi\|_3\|\nabla\varphi_h\|_0. \tag{4.87}$$

By use of (**??**), we have

$$\big(\nabla(R_h\psi - I_h\psi),\, \nabla\varphi_h\big) = \big(\nabla(\psi - I_h\psi),\, \nabla\varphi_h\big), \quad \forall\, \varphi_h \in X_h. \tag{4.88}$$

Taking $\varphi_h = R_h\psi - I_h\psi$ in (4.88) and using (4.87), we infer

$$\Big\|\nabla(R_h\psi - I_h\psi)\Big\|_0 \le Ch^2\|\psi\|_3. \tag{4.89}$$

As a result, from (4.87) and (4.89), we deduce

$$\big(\nabla(\psi - R_h\psi),\, \nabla\varphi_h\big) = \big(\nabla(\psi - I_h\psi),\, \nabla\varphi_h\big) + \big(\nabla(I_h\psi - R_h\psi),\, \nabla\varphi_h\big) \le Ch^2\|\psi\|_3\|\nabla\varphi_h\|_0. \tag{4.90}$$

The proof is completed. □

Based on Lemma 4.11, we can improve the estimate of E_7 in Lemma 4.2. In fact, E_7 can be split as

$$\begin{aligned} E_7 = &\Big(\big(e^j_{\tilde{c}_1} + (R_h\tilde{c}_1^j - \tilde{c}_1^j) + (\tilde{c}_1^j - P_0^K\tilde{c}_1^j)\big)\nabla(\psi^{j+1} - R_h\psi^{j+1}),\, \nabla e^{j+1}_{\tilde{c}_1}\Big) \\ &+ \sum_K P_0^K\tilde{c}_1^j\Big(\nabla(\psi^{j+1} - \tilde{R}_h\psi^{j+1}),\, \nabla e^{j+1}_{\tilde{c}_1}\Big), \end{aligned} \tag{4.91}$$

where $P_0^K\tilde{c}_1^j = \dfrac{1}{|K|}\displaystyle\int_K \tilde{c}_1^j\,\mathrm{d}x$.

Then, using Lemma 4.11, we infer

$$\begin{aligned} |E_7| &\le C\Big\|e^j_{\tilde{c}_1}\Big\|_{0,3}\Big\|\nabla(\psi^{j+1} - R_h\psi^{j+1})\Big\|_{0,6}\Big\|\nabla e^{j+1}_{\tilde{c}_1}\Big\|_0 + Ch^2\Big\|\psi^{j+1}\Big\|_3\Big\|\tilde{c}_1^j\Big\|_2\Big\|\nabla e^{j+1}_{\tilde{c}_1}\Big\|_0 \\ &\le CC_0^4h^4 + CC_0^2\Big\|e^j_{\tilde{c}_1}\Big\|_0^2 + \frac{1}{7}\Big\|\nabla e^{j+1}_{\tilde{c}_1}\Big\|_0^2. \end{aligned} \tag{4.92}$$

Consequently, we can improve the result of Theorem 4.4 as follows.

Theorem 4.12. *Under the conditions of Lemma 4.1, we have*

$$\Big\|c_1^{j+1} - c_{1h}^{j+1}\Big\|_0 + \Big\|c_2^{j+1} - c_{2h}^{j+1}\Big\|_0 + \Big\|u^{j+1} - u_h^{j+1}\Big\|_0 \le C_2\big(\tau + h^2\big). \tag{4.93}$$

Remark 4.13. Although only the first order spatial accuracy is reported in Theorem 4.10, a higher order finite element can also be applied, and higher order error estimates can be obtained under the condition of a higher regularity of solutions. In addition, on rectangular meshes, there has been some existing numerical work about the super-convergence analysis of the PNP equations and the Navier-Stokes equations [50, 36]. In a similar way, we can obtain the super-convergence results for the NSPNP equations, and the convergence order of the $\ell^\infty(H^1)$-norm can be improved to $\mathcal{O}(\tau + h^2)$.

5. Two physical properties

In this section, we aim to derive some important physical properties for the proposed scheme.

Theorem 5.1 (Global mass conservation)**.** *The finite element solutions $\big(c_{1h}^{j+1},\, c_{2h}^{j+1}\big)$ satisfy global mass conservation, that is,*

$$\int_\Omega c_{1h}^{j+1}\,\mathrm{d}x \;=\; \int_\Omega c_{2h}^{j+1}\,\mathrm{d}x \;=\; m_0\,|\Omega|. \tag{5.1}$$

The property (5.1) can be derived easily from (**??**).

Theorem 5.2 (Energy stability)**.** *Define a discrete energy by*

$$E\left(u_h^j,\, \psi_h^j\right) \;=\; \frac{1}{2}\left(\left\|u_h^j\right\|_0^2 + \left\|\nabla\psi_h^j\right\|_0^2\right). \tag{5.2}$$

Then, if τ and h are small enough, an energy stability

$$E\left(u_h^{j+1},\, \psi_h^{j+1}\right) \;\le\; E\left(u_h^j,\, \psi_h^j\right) \tag{5.3}$$

holds for $j = 0, 1, \dots, J-1$.

Proof. Step 1. Choose $\varphi_h = \Delta_h\psi_h^{j+1}$ and $\varphi_h = c_{1h}^{j+1} - c_{2h}^{j+1}$ in (3.2) respectively, and we get

$$\begin{aligned}\left\|\Delta_h\psi_h^{j+1}\right\|_0^2 &= \left(\nabla\psi_h^{j+1},\, \nabla\Delta_h\psi_h^{j+1}\right) = \left(c_{1h}^{j+1} - c_{2h}^{j+1},\, \Delta_h\psi_h^{j+1}\right)\\ &\le \left\|c_{1h}^{j+1} - c_{2h}^{j+1}\right\|_0 \left\|\Delta_h\psi_h^{j+1}\right\|_0,\end{aligned} \tag{5.4}$$

$$\begin{aligned}\left\|c_{1h}^{j+1} - c_{2h}^{j+1}\right\|_0^2 &= \left(\nabla\psi_h^{j+1},\, \nabla(c_{1h}^{j+1} - c_{2h}^{j+1})\right) = \left(c_{1h}^{j+1} - c_{2h}^{j+1},\, \Delta_h\psi_h^{j+1}\right)\\ &\le \left\|c_{1h}^{j+1} - c_{2h}^{j+1}\right\|_0 \left\|\Delta_h\psi_h^{j+1}\right\|_0,\end{aligned} \tag{5.5}$$

which implies

$$\left\|\Delta_h\psi_h^{j+1}\right\|_0 = \left\|c_{1h}^{j+1} - c_{2h}^{j+1}\right\|_0, \tag{5.6}$$

$$\left\|\Delta_h\psi_h^{j+1}\right\|_0^2 = \left(\nabla(c_{1h}^{j+1} - c_{2h}^{j+1}),\, \nabla\psi_h^{j+1}\right). \tag{5.7}$$

Step 2. Putting $v_h = u_h^{j+1}$ in (3.2) yields

$$\frac{1}{2\tau}\left(\left\|u_h^{j+1}\right\|_0^2 + \left\|u_h^{j+1} - u_h^j\right\|_0^2 - \left\|u_h^j\right\|_0^2\right) + \left\|\nabla u_h^{j+1}\right\|_0^2 = -\left((c_{1h}^j - c_{2h}^j)\nabla\psi_h^{j+1},\, u_h^{j+1}\right). \tag{5.8}$$

Step 3. From the last equation of (3.2), it is easy to find

$$\left(\nabla D_\tau\psi_h^{j+1},\, \nabla\varphi_h\right) = \left(D_\tau(c_{1h}^{j+1} - c_{2h}^{j+1}),\, \varphi_h\right), \quad \forall\, \varphi_h \in X_h. \tag{5.9}$$

Take $\eta_{1h} = \eta_{2h} = \psi_h^{j+1}$ in (3.2) and $\varphi_h = \psi_h^{j+1}$ in (5.9), respectively; then we arrive at

$$\begin{aligned}&\frac{1}{2\tau}\left(\left\|\nabla\psi_h^{j+1}\right\|_0^2 + \left\|\nabla(\psi_h^{j+1} - \psi_h^j)\right\|_0^2 - \left\|\nabla\psi_h^j\right\|_0^2\right) + \left(\nabla(c_{1h}^{j+1} - c_{2h}^{j+1}),\, \nabla\psi_h^{j+1}\right)\\ &\quad + \left((c_{1h}^j + c_{2h}^j)\nabla\psi_h^{j+1},\, \nabla\psi_h^{j+1}\right) = -b_2\left(u_h^j;\, c_{1h}^{j+1} - c_{2h}^{j+1},\, \psi_h^{j+1}\right).\end{aligned} \tag{5.10}$$

Step 4. Adding (5.8) and (5.10) and using (5.7), we deduce

$$\begin{aligned}&\frac{1}{2\tau}\left(\left\|u_h^{j+1}\right\|_0^2 + \left\|u_h^{j+1} - u_h^j\right\|_0^2 - \left\|u_h^j\right\|_0^2 + \left\|\nabla\psi_h^{j+1}\right\|_0^2 + \left\|\nabla(\psi_h^{j+1} - \psi_h^j)\right\|_0^2 - \left\|\nabla\psi_h^j\right\|_0^2\right)\\ &\quad + \left\|\nabla u_h^{j+1}\right\|_0^2 + \left\|\Delta_h\psi_h^{j+1}\right\|_0^2 + \left\|\sqrt{c_1^j + c_2^j}\,\nabla\psi_h^{j+1}\right\|_0^2\\ &= -\left((c_{1h}^j - c_1^j + c_{2h}^j - c_2^j)\nabla\psi_h^{j+1},\, \nabla\psi_h^{j+1}\right)\\ &\quad + \left((c_{1h}^{j+1} - c_{1h}^j - (c_{2h}^{j+1} - c_{2h}^j))\nabla\psi_h^{j+1},\, u_h^{j+1}\right)\\ &\quad - \left((c_{1h}^{j+1} - c_{2h}^{j+1})\nabla\psi_h^{j+1},\, u_h^{j+1}\right)\\ &\quad + b_2\left(u_h^{j+1} - u_h^j;\, c_{1h}^{j+1} - c_{2h}^{j+1},\, \psi_h^{j+1}\right) - b_2\left(u_h^{j+1};\, c_{1h}^{j+1} - c_{2h}^{j+1},\, \psi_h^{j+1}\right)\\ &=: \sum_{i=1}^{5} R_i.\end{aligned} \tag{5.11}$$

Next, we estimate the above five terms. By use of Theorem **??**, it is easy to check

$$\left\|c_{1h}^{j+1}-c_{1h}^{j}\right\|_0 \le \left\|c_{1h}^{j+1}-c_{1}^{j+1}\right\|_0 + \left\|c_{1h}^{j}-c_{1}^{j}\right\|_0 + \left\|c_{1}^{j+1}-c_{1}^{j}\right\|_0 \le C_2(\tau+h). \tag{5.12}$$

Similarly, we obtain

$$\left\|c_{2h}^{j+1}-c_{2h}^{j}\right\|_0 \le C_2(\tau+h), \qquad \left\|\nabla\left(u_h^{j+1}-u_h^{j}\right)\right\|_0 \le C_5(\tau+h). \tag{5.13}$$

Then, based on (5.6), (5.12), (5.13) and Theorem **??**, we infer

$$\begin{aligned}
|R_1|+|R_2|+|R_4| \le{}& \left(\left\|c_{1h}^{j}-c_{1}^{j}\right\|_0 + \left\|c_{2h}^{j}-c_{2}^{j}\right\|_0\right)\left\|\nabla\psi_h^{j+1}\right\|_{0,3}\left\|\nabla\psi_h^{j+1}\right\|_{0,6}\\
&+\left(\left\|c_{1h}^{j+1}-c_{1h}^{j}\right\|_0 + \left\|c_{2h}^{j+1}-c_{2h}^{j}\right\|_0\right)\left\|\nabla\psi_h^{j+1}\right\|_{0,3}\left\|u_h^{j+1}\right\|_{0,6}\\
&+\left\|\nabla\left(u_h^{j+1}-u_h^{j}\right)\right\|_0\left\|c_{1h}^{j+1}-c_{2h}^{j+1}\right\|_0\left\|\Delta_h\psi_h^{j+1}\right\|_0\\
\le{}& C_5(\tau+h)\left\|\Delta_h\psi_h^{j+1}\right\|_0^2 + \frac{1}{2}\left\|\nabla u_h^{j+1}\right\|_0^2.
\end{aligned} \tag{5.14}$$

Using Green's formula, Lemma 2.1 (equation (**??**)), Theorem **??** and (5.6), we get

$$\begin{aligned}
|R_3+R_5| &= \frac{1}{2}\left|\left(\operatorname{div} u_h^{j+1},\ (c_{1h}^{j+1}-c_{2h}^{j+1})\psi_h^{j+1}\right)\right|\\
&= \frac{1}{2}\left|\left(\operatorname{div}(u_h^{j+1}-u^{j+1}),\ (c_{1h}^{j+1}-c_{2h}^{j+1})\psi_h^{j+1}\right)\right|\\
&\le C_5(\tau+h)\left\|c_{1h}^{j+1}-c_{2h}^{j+1}\right\|_0\left\|\psi_h^{j+1}\right\|_{0,\infty} \le C_5(\tau+h)\left\|\Delta_h\psi_h^{j+1}\right\|_0^2.
\end{aligned} \tag{5.15}$$

Inserting (5.14) and (5.15) into (5.11) yields

$$\begin{aligned}
&\frac{1}{2}\left(\left\|u_h^{j+1}\right\|_0^2 + \left\|\nabla\psi_h^{j+1}\right\|_0^2\right) + \frac{1}{2}\left(\left\|u_h^{j+1}-u_h^{j}\right\|_0^2 + \left\|\nabla(\psi_h^{j+1}-\psi_h^{j})\right\|_0^2\right)\\
&\quad+\tau\left(1-C_5(\tau+h)\right)\left\|\Delta_h\psi_h^{j+1}\right\|_0^2 + \frac{1}{2}\tau\left\|\nabla u_h^{j+1}\right\|_0^2 + \tau\left\|\sqrt{c_1^j+c_2^j}\,\nabla\psi_h^{j+1}\right\|_0^2\\
&\le \frac{1}{2}\left(\left\|u_h^{j}\right\|_0^2 + \left\|\nabla\psi_h^{j}\right\|_0^2\right).
\end{aligned} \tag{5.16}$$

If τ and h are small enough, (5.3) holds. □

6. Numerical implementation

In this section, we present three numerical examples to verify the theoretical results. The first two examples have smooth exact solutions, which are used to confirm the convergence orders. The third example is to show the stability of the scheme.

Example 6.1. The first example refers to [15] with some modifications. Take $\Omega = [0,1]\times[0,1]$, and consider the example with following exact solutions

$$u_1 = -e^{-t}\sin^2(\pi x_1)\sin(\pi x_2)\cos(\pi x_2), \tag{6.1}$$

$$u_2 = \quad e^{-t}\sin(\pi x_1)\cos(\pi x_1)\sin^2(\pi x_2), \tag{6.2}$$

$$p = \sin(\pi x_1)\cos(\pi x_2), \tag{6.3}$$

$$c_1 = \frac{1}{2}e^{-t}\left(\cos(\pi x_1)\cos(\pi x_2)+1\right), \tag{6.4}$$

$$c_2 = \frac{1}{2}e^{-t}\left(\cos(2\pi x_1)\cos(2\pi x_2)+1\right), \tag{6.5}$$

$$\psi = \frac{1}{2\pi^2}e^{-t}\left(\cos(\pi x_1)\cos(\pi x_2) - \frac{1}{4}\cos(2\pi x_1)\cos(2\pi x_2)\right). \tag{6.6}$$

We use two pairs of elements in the computation. The first pair (Pair I) employs the linear element to approximate c_1, c_2, ψ, and the MINI element for u and p. The second pair (Pair II) uses the rectangular bilinear/Q_1-Q_0 element described in Section **??**. The errors and convergence orders at $T = 1$ are shown in Tables 1–8.

Table 1
Numerical results of Pair I for c_1 (Example 6.1).

h	$\|c_1^j - c_{1h}^j\|_0$	order	$\|\nabla(c_1^j - c_{1h}^j)\|_0$	order
$\frac{1}{4}\times\frac{1}{4}$	1.71216×10^{-2}	—	1.49070×10^{-1}	—
$\frac{1}{8}\times\frac{1}{8}$	4.72101×10^{-3}	1.85865	7.84503×10^{-2}	0.92613
$\frac{1}{16}\times\frac{1}{16}$	1.21555×10^{-3}	1.95749	3.98565×10^{-2}	0.97696
$\frac{1}{24}\times\frac{1}{24}$	5.43558×10^{-4}	1.98492	2.66612×10^{-2}	0.99165
$\frac{1}{32}\times\frac{1}{32}$	3.06444×10^{-4}	1.99213	2.00212×10^{-2}	0.99559

Table 2
Numerical results of Pair I for c_2 (Example 6.1).

h	$\|c_2^j - c_{2h}^j\|_0$	order	$\|\nabla(c_2^j - c_{2h}^j)\|_0$	order
$\frac{1}{4}\times\frac{1}{4}$	4.81350×10^{-2}	—	5.26501×10^{-1}	—
$\frac{1}{8}\times\frac{1}{8}$	1.59385×10^{-2}	1.59457	3.02892×10^{-1}	0.79763
$\frac{1}{16}\times\frac{1}{16}$	4.32229×10^{-3}	1.88264	1.57958×10^{-1}	0.93926
$\frac{1}{24}\times\frac{1}{24}$	1.95245×10^{-3}	1.95998	1.06202×10^{-1}	0.97908
$\frac{1}{32}\times\frac{1}{32}$	1.10471×10^{-3}	1.97962	7.98971×10^{-2}	0.98930

Table 3
Numerical results of Pair I for ψ (Example 6.1).

h	$\|\psi^j - \psi_h^j\|_0$	order	$\|\nabla(\psi^j - \psi_h^j)\|_0$	order
$\frac{1}{4}\times\frac{1}{4}$	2.13515×10^{-3}	—	1.30796×10^{-2}	—
$\frac{1}{8}\times\frac{1}{8}$	7.06763×10^{-4}	1.59504	6.67585×10^{-3}	0.97030
$\frac{1}{16}\times\frac{1}{16}$	1.93532×10^{-4}	1.86866	3.30018×10^{-3}	1.01641
$\frac{1}{24}\times\frac{1}{24}$	8.76128×10^{-5}	1.95458	2.19194×10^{-3}	1.00918
$\frac{1}{32}\times\frac{1}{32}$	4.96106×10^{-5}	1.97692	1.64162×10^{-3}	1.00495

Table 4
Numerical results of Pair I for u (Example 6.1).

h	$\|u^j - u_h^j\|_0$	order	$\|\nabla(u^j - u_h^j)\|_0$	order
$\frac{1}{4}\times\frac{1}{4}$	4.58742×10^{-2}	—	4.83426×10^{-1}	—
$\frac{1}{8}\times\frac{1}{8}$	1.30885×10^{-2}	1.80939	2.52312×10^{-1}	0.93809
$\frac{1}{16}\times\frac{1}{16}$	3.31031×10^{-3}	1.98326	1.26768×10^{-1}	0.99302
$\frac{1}{24}\times\frac{1}{24}$	1.46916×10^{-3}	2.00350	8.44530×10^{-2}	1.00172
$\frac{1}{32}\times\frac{1}{32}$	8.25074×10^{-4}	2.00560	6.32890×10^{-2}	1.00279

From Tables 1–8, we can see that the $l^\infty(L^2)$ and $l^\infty(H^1)$-norm errors of all variables are convergent at rates of $\mathcal{O}(h^2)$ and $\mathcal{O}(h)$ respectively for the two finite element pairs. It is clear that all variables achieve optimal convergence order, which accords with our theoretical analysis completely.

In addition, it seems that the errors of the approximate solutions obtained using Pair II are less than those of Pair I. Moreover, Pair II has fewer degrees of freedom than Pair I, which means less computation time. However, Pair II is the rectangular element and Pair I is the triangular element, so the latter has better adaptability for the computational domain.

Example 6.2. The second numerical example is provided in three-dimensional space from [15, 35], which has the following exact solutions

$$\begin{aligned}
u_1 &= -e^{-t}\sin(2\pi x_2)\sin(2\pi x_3)\big(\cos(2\pi x_1)-1\big),\\
u_2 &= -e^{-t}\sin(2\pi x_1)\sin(2\pi x_3)\big(\cos(2\pi x_2)-1\big),\\
u_3 &= \quad 2e^{-t}\sin(2\pi x_1)\sin(2\pi x_2)\big(\cos(2\pi x_3)-1\big),\\
p &= e^{-t}\big(\sin(2\pi x_1)+\sin(2\pi x_2)+\sin(2\pi x_3)\big),\\
c_1 &= 3\pi^2 e^{-t}\cos(\pi x_1)\cos(\pi x_2)\cos(\pi x_3),\\
c_2 &= \pi^2 t^3\cos(\pi x_3),\\
\psi &= e^{-t}\cos(\pi x_1)\cos(\pi x_2)\cos(\pi x_3) - t^3\cos(\pi x_3).
\end{aligned} \tag{6.7}$$

Table 5
Numerical results of Pair II for c_1 (Example 6.1).

h	$\|c_1^j - c_{1h}^j\|_0$	order	$\|\nabla(c_1^j - c_{1h}^j)\|_0$	order
$\frac{1}{4}\times\frac{1}{4}$	1.13112×10^{-2}	—	9.22339×10^{-2}	—
$\frac{1}{8}\times\frac{1}{8}$	2.84991×10^{-3}	1.98876	4.62650×10^{-2}	0.99538
$\frac{1}{16}\times\frac{1}{16}$	7.13927×10^{-4}	1.99707	2.31534×10^{-2}	0.99870
$\frac{1}{24}\times\frac{1}{24}$	3.17422×10^{-4}	1.99906	1.54383×10^{-2}	0.99958
$\frac{1}{32}\times\frac{1}{32}$	1.78573×10^{-4}	1.99954	1.15794×10^{-2}	0.99979

Table 6
Numerical results of Pair II for c_2 (Example 6.1).

h	$\|c_2^j - c_{2h}^j\|_0$	order	$\|\nabla(c_2^j - c_{2h}^j)\|_0$	order
$\frac{1}{4}\times\frac{1}{4}$	2.45008×10^{-2}	—	3.66530×10^{-1}	—
$\frac{1}{8}\times\frac{1}{8}$	6.14041×10^{-3}	1.99642	1.84443×10^{-1}	0.99075
$\frac{1}{16}\times\frac{1}{16}$	1.53689×10^{-3}	1.99832	9.25268×10^{-2}	0.99524
$\frac{1}{24}\times\frac{1}{24}$	6.83240×10^{-4}	1.99936	6.17268×10^{-2}	0.99831
$\frac{1}{32}\times\frac{1}{32}$	3.84358×10^{-4}	1.99967	4.63064×10^{-2}	0.99915

Table 7
Numerical results of Pair II for ψ (Example 6.1).

h	$\|\psi^j - \psi_h^j\|_0$	order	$\|\nabla(\psi^j - \psi_h^j)\|_0$	order
$\frac{1}{4}\times\frac{1}{4}$	6.44562×10^{-4}	—	6.86706×10^{-3}	—
$\frac{1}{8}\times\frac{1}{8}$	1.75746×10^{-4}	1.87483	3.35922×10^{-3}	1.03157
$\frac{1}{16}\times\frac{1}{16}$	4.48532×10^{-5}	1.97021	1.66481×10^{-3}	1.01277
$\frac{1}{24}\times\frac{1}{24}$	2.00116×10^{-5}	1.99052	1.10787×10^{-3}	1.00445
$\frac{1}{32}\times\frac{1}{32}$	1.12717×10^{-5}	1.99531	8.30366×10^{-4}	1.00224

Table 8
Numerical results of Pair II for u (Example 6.1).

h	$\|u^j - u_h^j\|_0$	order	$\|\nabla(u^j - u_h^j)\|_0$	order
$\frac{1}{4}\times\frac{1}{4}$	2.52809×10^{-2}	—	3.69245×10^{-1}	—
$\frac{1}{8}\times\frac{1}{8}$	6.34426×10^{-3}	1.99452	1.85081×10^{-1}	0.99642
$\frac{1}{16}\times\frac{1}{16}$	1.58665×10^{-3}	1.99947	9.26102×10^{-2}	0.99892
$\frac{1}{24}\times\frac{1}{24}$	7.05211×10^{-4}	1.99989	6.17515×10^{-2}	0.99954
$\frac{1}{32}\times\frac{1}{32}$	3.96687×10^{-4}	1.99995	4.63169×10^{-2}	0.99976

In the computation, we take $\Omega = [0,1]^3$ and use a uniform tetrahedral partition in each direction. The errors and convergence orders at $T = 1$ are listed in Tables 9–12, which show that all the error estimates achieve optimal convergence order, as in the two-dimensional case.

Example 6.3. The third example is taken from [44, 15, 8]. Set $\Omega = [0,1]\times[0,1]$, and the initial values are given as follows

$$c_{10} = \begin{cases} 1, & (0,1)^2 \setminus \left\{(0,0.75)\times(0,1)\cup(0.75,1)\times\left(0,\frac{11}{20}\right)\right\}, \\ 10^{-6}, & \text{otherwise}, \end{cases} \tag{6.8}$$

$$c_{20} = \begin{cases} 1, & (0,1)^2 \setminus \left\{(0,0.75)\times(0,1)\cup(0.75,1)\times\left(\frac{9}{20},1\right)\right\}, \\ 10^{-6}, & \text{otherwise}, \end{cases} \tag{6.9}$$

and $u_0 = \mathbf{0}$.

In the computations, we set the time step $\tau = 0.001$, the mesh size $h = 1/40$, and use Pair I. The simulation results at $t = 0.002$, $t = 0.02$ and $t = 0.1$ are shown in Figs. 1 and 2, which match the results in [44, 15, 8]. Moreover, from Fig. 3, we can see that the total energy is decreasing and the global mass conservation is preserved during the evolution, which confirms numerically our findings in Theorems 5.1 and 5.2.

Table 9
Errors and convergence orders of Pair I for c_1 (Example 6.2).

h	$\|c_1^j - c_{1h}^j\|_0$	order	$\|\nabla(c_1^j - c_{1h}^j)\|_0$	order
$\frac{1}{4}\times\frac{1}{4}\times\frac{1}{4}$	8.86014×10^{-1}	—	$9.53915\times10^{+0}$	—
$\frac{1}{8}\times\frac{1}{8}\times\frac{1}{8}$	2.69385×10^{-1}	1.71766	$5.14715\times10^{+0}$	0.89009
$\frac{1}{12}\times\frac{1}{12}\times\frac{1}{12}$	1.24351×10^{-1}	1.90652	$3.46474\times10^{+0}$	0.97618
$\frac{1}{16}\times\frac{1}{16}\times\frac{1}{16}$	7.16936×10^{-2}	1.91430	$2.63284\times10^{+0}$	0.95444
$\frac{1}{20}\times\frac{1}{20}\times\frac{1}{20}$	4.61452×10^{-2}	1.97455	$2.10730\times10^{+0}$	0.99781

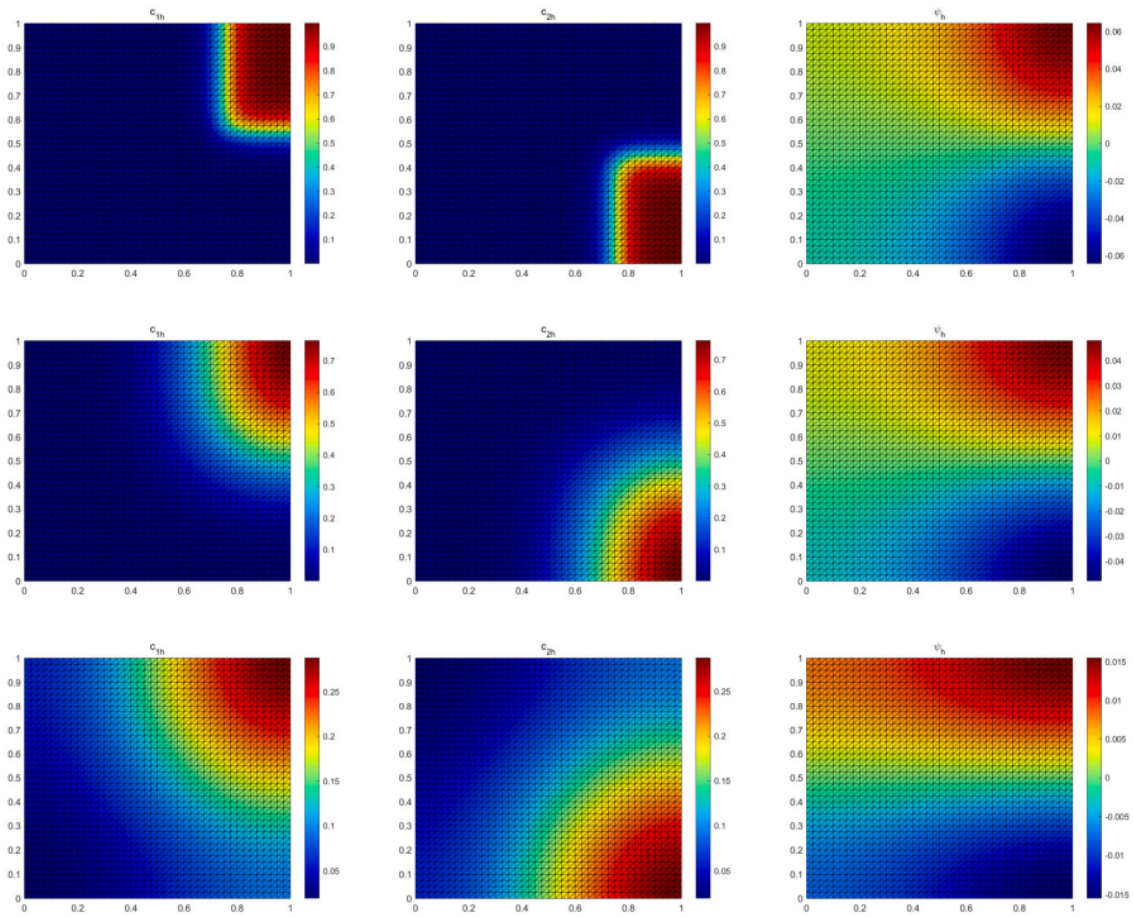

Figure 1: Snapshots of the approximate solutions (c_{1h}, c_{2h}, ψ_h) at $t = 0.002$ (top row), $t = 0.02$ (middle) and $t = 0.1$ (bottom).

Table 10
Errors and convergence orders of Pair I for c_2 (Example 6.2).

h	$\|c_2^j - c_{2h}^j\|_0$	order	$\|\nabla(c_2^j - c_{2h}^j)\|_0$	order
$\frac{1}{4}\times\frac{1}{4}\times\frac{1}{4}$	3.41539×10^{-1}	—	$5.14710\times10^{+0}$	—
$\frac{1}{8}\times\frac{1}{8}\times\frac{1}{8}$	9.37020×10^{-2}	1.86590	$2.52773\times10^{+0}$	1.02592
$\frac{1}{12}\times\frac{1}{12}\times\frac{1}{12}$	4.37079×10^{-2}	1.88078	$1.70781\times10^{+0}$	0.96707
$\frac{1}{16}\times\frac{1}{16}\times\frac{1}{16}$	2.41287×10^{-2}	2.06522	$1.24973\times10^{+0}$	1.08550
$\frac{1}{20}\times\frac{1}{20}\times\frac{1}{20}$	1.56550×10^{-2}	1.93872	$1.00551\times10^{+0}$	0.97443

7. Conclusion and future work

In this work, we present a linearized backward Euler finite element scheme for the NSPNP equations, and prove the existence and uniqueness of this scheme. We then obtain the unconditional error estimates of all variables in $l^\infty(L^2)$ and $l^\infty(H^1)$ norms. Moreover, we give two important physical properties: the global mass conservation and energy stability.

However, the concentration positivity cannot be theoretically justified, which is also important for the numerical scheme of the NSPNP equations. In fact, for the pure PNP equations, there has been some existing numerical

Table 11
Errors and convergence orders of Pair I for ψ (Example 6.2).

h	$\|\psi^j - \psi_h^j\|_0$	order	$\|\nabla(\psi^j - \psi_h^j)\|_0$	order
$\frac{1}{4} \times \frac{1}{4} \times \frac{1}{4}$	5.13316×10^{-2}	—	6.03225×10^{-1}	—
$\frac{1}{8} \times \frac{1}{8} \times \frac{1}{8}$	1.40437×10^{-2}	1.86992	3.07475×10^{-1}	0.97223
$\frac{1}{12} \times \frac{1}{12} \times \frac{1}{12}$	6.46538×10^{-3}	1.91315	2.08062×10^{-1}	0.96323
$\frac{1}{16} \times \frac{1}{16} \times \frac{1}{16}$	3.63972×10^{-3}	1.99719	1.54403×10^{-1}	1.03681
$\frac{1}{20} \times \frac{1}{20} \times \frac{1}{20}$	2.34852×10^{-3}	1.96342	1.24095×10^{-1}	0.97928

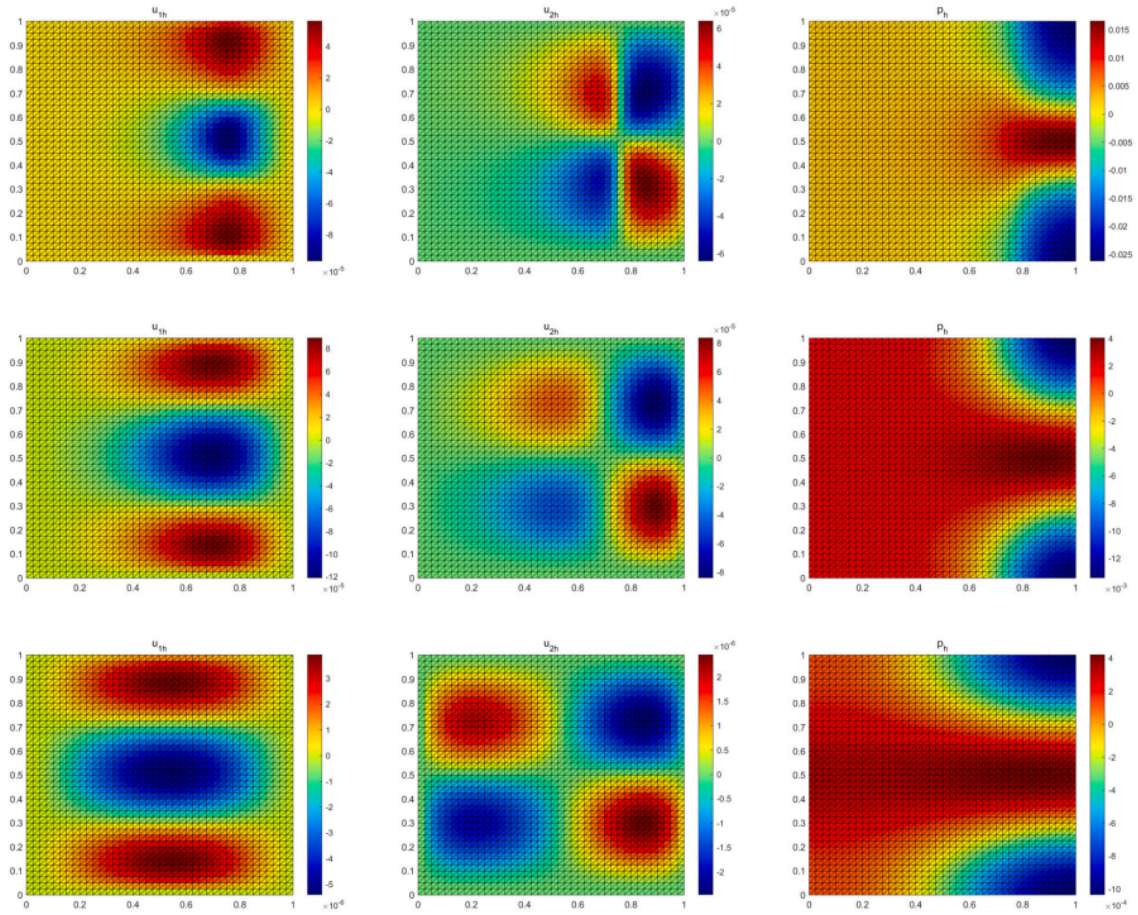

Figure 2: Snapshots of the approximate solutions (u_{1h}, u_{2h}, p_h) at $t = 0.002$ (top row), $t = 0.02$ (middle) and $t = 0.1$ (bottom).

Table 12
Errors and convergence orders of Pair I for u (Example 6.2).

h	$\|u^j - u_h^j\|_0$	order	$\|\nabla(u^j - u_h^j)\|_0$	order
$\frac{1}{4} \times \frac{1}{4} \times \frac{1}{4}$	3.27914×10^{-1}	—	$3.95123 \times 10^{+0}$	—
$\frac{1}{8} \times \frac{1}{8} \times \frac{1}{8}$	1.14243×10^{-1}	1.52121	$2.24376 \times 10^{+0}$	0.81638
$\frac{1}{12} \times \frac{1}{12} \times \frac{1}{12}$	5.35706×10^{-2}	1.86780	$1.51920 \times 10^{+0}$	0.96178
$\frac{1}{16} \times \frac{1}{16} \times \frac{1}{16}$	3.09728×10^{-2}	1.90450	$1.15613 \times 10^{+0}$	0.94933
$\frac{1}{20} \times \frac{1}{20} \times \frac{1}{20}$	1.99553×10^{-2}	1.97010	9.25695×10^{-1}	0.99616

methods which keep these physical properties. In the future, we will try to apply these methods to the NSPNP equations. In addition, our numerical scheme is only first-order accurate in time, and in a forthcoming research we plan to propose some second-order accurate numerical schemes for the NSPNP equations, such as the Crank–Nicolson scheme and the BDF2 scheme. Finally, it is interesting and challenging to construct robust schemes to solve some practical problems, such as charging dynamics in a nanopore, electroosmotic flow in a nanochannel, and electroconvective flow on an ion-selective surface [54], and so on.

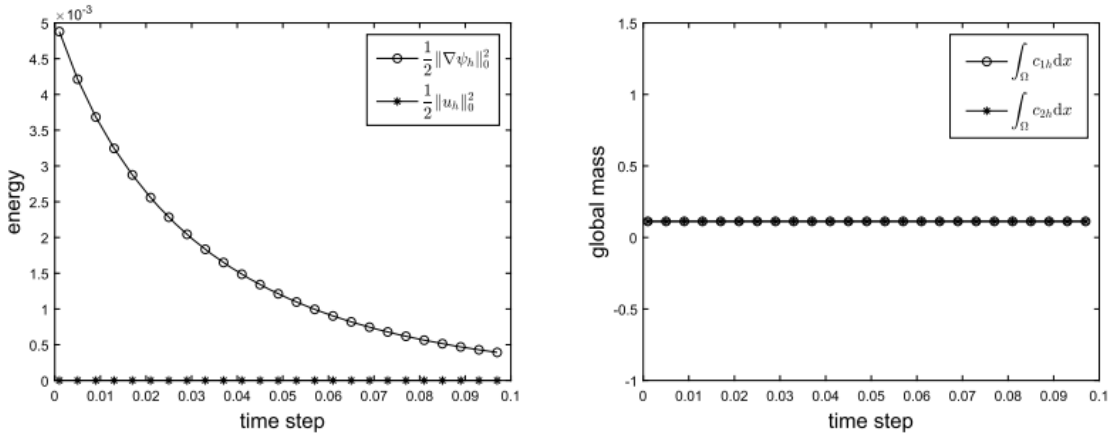

Figure 3: Evolution of the energy (left) and global mass (right).

References

[1] Bothe, D., Fischer, A., Saal, J., 2014. Global well-posedness and stability of electrokinetic flows. SIAM J. Math. Anal. 46, 1263–1316.

[2] Bousquet, A., Hu, X., Metti, M.S., et al., 2018. Newton solvers for drift-diffusion and electrokinetic equations. SIAM J. Sci. Comput. 40, B982–B1006.

[3] Chen, W., Feng, W., Liu, Y., et al., 2019. A second order energy stable scheme for the Cahn-Hilliard-Hele-Shaw equations. Discrete Contin. Dyn. B 24, 149–182.

[4] Chen, W., Liu, Y., Wang, C., et al., 2016. Convergence analysis of a fully discrete finite difference scheme for the Cahn-Hilliard-Hele-Shaw equation. Math. Comput. 85, 2231–2257.

[5] Constantin, P., Ignatova, M., 2019. On the Nernst-Planck-Navier-Stokes system. Arch. Ration. Mech. Anal. 232, 1379–1428.

[6] Constantin, P., Ignatova, M., Lee, F.N., 2021. Nernst-Planck-Navier-Stokes systems far from equilibrium. Arch. Ration. Mech. Anal. 240, 1147–1168.

[7] Dehghan, M., Gharibi, Z., 2022. An analysis of weak Galerkin finite element method for a steady state Boussinesq problem. J. Comput. Appl. Math. 406, 114029.

[8] Dehghan, M., Gharibi, Z., Ruiz-Baier, R., 2023. Optimal error estimates of coupled and divergence-free virtual element methods for the Poisson-Nernst-Planck/Navier-Stokes equations and applications in electrochemical systems. J. Sci. Comput. 94, 72.

[9] Dehghan, M., Hooshyarfarzin, B., Abbaszadeh, M., 2022. Numerical simulation based on a combination of finite element method and proper orthogonal decomposition to prevent the groundwater contamination. Eng. Comput. 38, 3445–3461.

[10] Diegel, A., Wang, C., Wang, X., et al., 2017. Convergence analysis and error estimates for a second order accurate finite element method for the Cahn-Hilliard-Navier-Stokes system. Numer. Math. 137, 495–534.

[11] Ding, J., Wang, Z., Zhou, S., 2020. Structure-preserving and efficient numerical methods for ion transport. J. Comput. Phys. 418, 109597.

[12] Fischer, A., Saal, J., 2017. Global weak solutions in three space dimensions for electrokinetic flow processes. J. Evol. Equ. 17, 309–333.

[13] Flavell, A., Machen, M., Eisenberg, B., et al., 2014. A conservative finite difference scheme for Poisson-Nernst-Planck equations. J. Comput. Electron. 13, 235–249.

[14] Fu, G., Xu, Z., 2022. High-order space-time finite element methods for the Poisson-Nernst-Planck equations: positivity and unconditional energy stability. Comput. Methods Appl. Mech. Eng. 395, 115031.

[15] Gao, H., He, D., 2017. Linearized conservative finite element methods for the Nernst-Planck-Poisson equations. J. Sci. Comput. 72, 1269–1289.

[16] Gao, H., Sun, P., 2018. A linearized local conservative mixed finite element method for Poisson-Nernst-Planck equations. J. Sci. Comput. 77, 793–817.

[17] Gharibi, Z., Dehghan, M., 2021. Convergence analysis of weak Galerkin flux-based mixed finite element method for solving singularly perturbed convection-diffusion-reaction problem. Appl. Numer. Math. 163, 303–316.

[18] Gharibi, Z., Dehghan, M., Abbaszadeh, M., 2021. Numerical analysis of locally conservative weak Galerkin dual-mixed finite element method for the time-dependent Poisson-Nernst-Planck system. Comput. Math. Appl. 92, 88–108.

[19] Gong, H., Wang, C., Zhang, X., 2021. Partial regularity of suitable weak solutions of the Navier-Stokes-Planck-Nernst-Poisson equation. SIAM J. Math. Anal. 53, 3306–3337.

[20] Gubbiotti, A., Baldelli, M., Muccio, G.D., et al., 2022. Electroosmosis in nanopores: computational methods and technological applications. Adv. Phys. X 7, 2036638.

[21] Hao, T., Ma, M., Xu, X., 2022. Adaptive finite element approximation for steady-state Poisson-Nernst-Planck equations. Adv. Comput. Math. 48, 1–27.

[22] He, D., Pan, K., Yue, X., 2019. A positivity preserving and free energy dissipative difference scheme for the Poisson-Nernst-Planck system. J. Sci. Comput. 81, 436–458.

[23] He, M., Sun, P., 2017. Error analysis of mixed finite element method for Poisson-Nernst-Planck system. Numer. Methods Partial Differ. Equ. 33, 1924–1948.

[24] He, M., Sun, P., 2018. Mixed finite element analysis for the Poisson-Nernst-Planck/Stokes coupling. J. Comput. Appl. Math. 341, 61–79.

[25] He, M., Sun, P., 2021. Mixed finite element method for modified Poisson-Nernst-Planck/Navier-Stokes equations. J. Sci. Comput. 87, 1–33.

[26] Heywood, J.G., Rannacher, R., 1982. Finite element approximation of the nonstationary Navier-Stokes problem. I. Regularity of solutions and second-order error estimates for spatial discretization. SIAM J. Numer. Anal. 19, 275–311.

[27] Heywood, J.G., Rannacher, R., 1990. Finite-element approximation of the nonstationary Navier-Stokes problem. Part IV: Error analysis for second-order time discretization. SIAM J. Numer. Anal. 27, 353–384.

[28] Huang, F., Shen, J., 2021. Bound/positivity preserving and energy stable scalar auxiliary variable schemes for dissipative systems: applications to Keller-Segel and Poisson-Nernst-Planck equations. SIAM J. Sci. Comput. 43, A1832–A1857.
[29] Jerome, J.W., Sacco, R., 2009. Global weak solutions for an incompressible charged fluid with multi-scale couplings: initial-boundary-value problem. Nonlinear Anal. 71, e2487–e2497.
[30] John, V., 2016. Finite Element Methods for Incompressible Flow Problems. Springer-Verlag, Berlin.
[31] Kirby, B.J., 2010. Micro- and Nanoscale Fluid Mechanics: Transport in Microfluidic Devices. Cambridge University Press, New York.
[32] Li, J., Ying, J., 2022. A stabilized finite volume element method for solving Poisson-Nernst-Planck equations. Int. J. Numer. Methods Biomed. Eng. 38, e3543.
[33] Li, M., Xiao, L., Li, Z., 2023a. The supercloseness property of the Stoke projection for the transient Navier-Stokes equations and global superconvergence analysis. J. Appl. Math. Comput. 69, 2041–2057.
[34] Li, Z., Li, M., Shi, D., 2021. Unconditional convergence and superconvergence analysis for the transient Stokes equations with damping. Appl. Math. Comput. 389, 125572.
[35] Li, Z., Xiao, L., Li, M., et al., 2023b. Error estimates for the finite element method of the chemotaxis-Navier-Stokes equations. J. Appl. Math. Comput. 69, 1–27.
[36] Lin, Q., Lin, J., 2006. Finite Element Methods: Accuracy and Improvement. Science Press, Beijing.
[37] Liu, C., Wang, C., Wise, S., et al., 2021. A positivity-preserving, energy stable and convergent numerical scheme for the Poisson-Nernst-Planck system. Math. Comput. 90, 2071–2106.
[38] Liu, C., Wang, C., Wise, S.M., et al., 2022. An iteration solver for the Poisson-Nernst-Planck system and its convergence analysis. J. Comput. Appl. Math. 406, 114017.
[39] Liu, H., Maimaitiyiming, W., 2021. Efficient, positive, and energy stable schemes for multi-d Poisson-Nernst-Planck systems. J. Sci. Comput. 87, 1–36.
[40] Liu, X., Xu, C., 2017. Efficient time-stepping/spectral methods for the Navier-Stokes-Nernst-Planck-Poisson equations. Commun. Comput. Phys. 21, 1408–1428.
[41] Liu, Y., Chen, W., Wang, C., et al., 2017. Error analysis of a mixed finite element method for a Cahn-Hilliard-Hele-Shaw system. Numer. Math. 135, 679–709.
[42] Markowich, P.A., Ringhofer, C.A., Schmeiser, C., 1990. Semiconductor Equations. Springer-Verlag, New York.
[43] Metti, M.S., Xu, J., Liu, C., 2016. Energetically stable discretizations for charge transport and electrokinetic models. J. Comput. Phys. 306, 1–18.
[44] Prohl, A., Schmuck, M., 2009. Convergent discretizations for the Nernst-Planck-Poisson system. Numer. Math. 111, 591–630.
[45] Prohl, A., Schmuck, M., 2010. Convergent finite element discretizations of the Navier-Stokes-Nernst-Planck-Poisson system. ESAIM: Math. Model. Numer. Anal. 44, 531–571.
[46] Ramos, A., 2011. Electrokinetics and Electrohydrodynamics in Microsystems. Springer, New York.
[47] Schmuck, M., 2009. Analysis of the Navier-Stokes-Nernst-Planck-Poisson system. Math. Models Methods Appl. Sci. 19, 993–1014.
[48] Shen, J., Xu, J., 2021. Unconditionally positivity preserving and energy dissipative schemes for Poisson-Nernst-Planck equations. Numer. Math. 148, 671–697.
[49] Shen, R., Shu, S., Yang, Y., et al., 2020. A decoupling two-grid method for the time-dependent Poisson-Nernst-Planck equations. Numer. Algorithms 83, 1613–1651.
[50] Shi, D., Yang, H., 2019. Superconvergence analysis of finite element method for Poisson-Nernst-Planck equations. Numer. Methods Partial Differ. Equ. 35, 1206–1223.
[51] Shi, X., Lu, L., 2021. Nonconforming finite element method for coupled Poisson-Nernst-Planck equations. Numer. Methods Partial Differ. Equ. 37, 2714–2729.
[52] Su, S., Tang, H., 2022. A positivity-preserving and free energy dissipative hybrid scheme for the Poisson-Nernst-Planck equations on polygonal and polyhedral meshes. Comput. Math. Appl. 108, 33–48.
[53] Sun, Y., Sun, P., Zheng, B., et al., 2016. Error analysis of finite element method for Poisson-Nernst-Planck equations. J. Comput. Appl. Math. 301, 28–43.
[54] Wang, C., Bao, J., Pan, W., et al., 2017. Modeling electrokinetics in ionic liquids. Electrophoresis , 1–13.
[55] Wang, Q., Li, H., Zhang, L., et al., 2021. A stabilized finite element method for the Poisson-Nernst-Planck equations in three-dimensional ion channel simulations. Appl. Math. Lett. 111, 106652.
[56] Xie, D., Lu, B., 2020. An effective finite element iterative solver for a Poisson-Nernst-Planck ion channel model with periodic boundary conditions. SIAM J. Sci. Comput. 42, B1490–B1516.
[57] Yang, Y., Liu, Y., Shu, S., 2022a. Error analysis of virtual element methods for the time-dependent Poisson-Nernst-Planck equations. arXiv preprint arXiv:2207.07231 .
[58] Yang, Y., Shen, R., Fang, M., et al., 2022b. Local averaging type a posteriori error estimates for the nonlinear steady-state Poisson-Nernst-Planck equations. J. Comput. Appl. Math. 404, 113874.
[59] Ying, J., Fan, R., Li, J., et al., 2021. A new block preconditioner and improved finite element solver of Poisson-Nernst-Planck equation. J. Comput. Phys. 430, 110098.
[60] Zhu, W., Yang, Y., Ji, G., et al., 2022. Residual type a posteriori error estimates for the time-dependent Poisson-Nernst-Planck equations. J. Sci. Comput. 90, 1–35.